

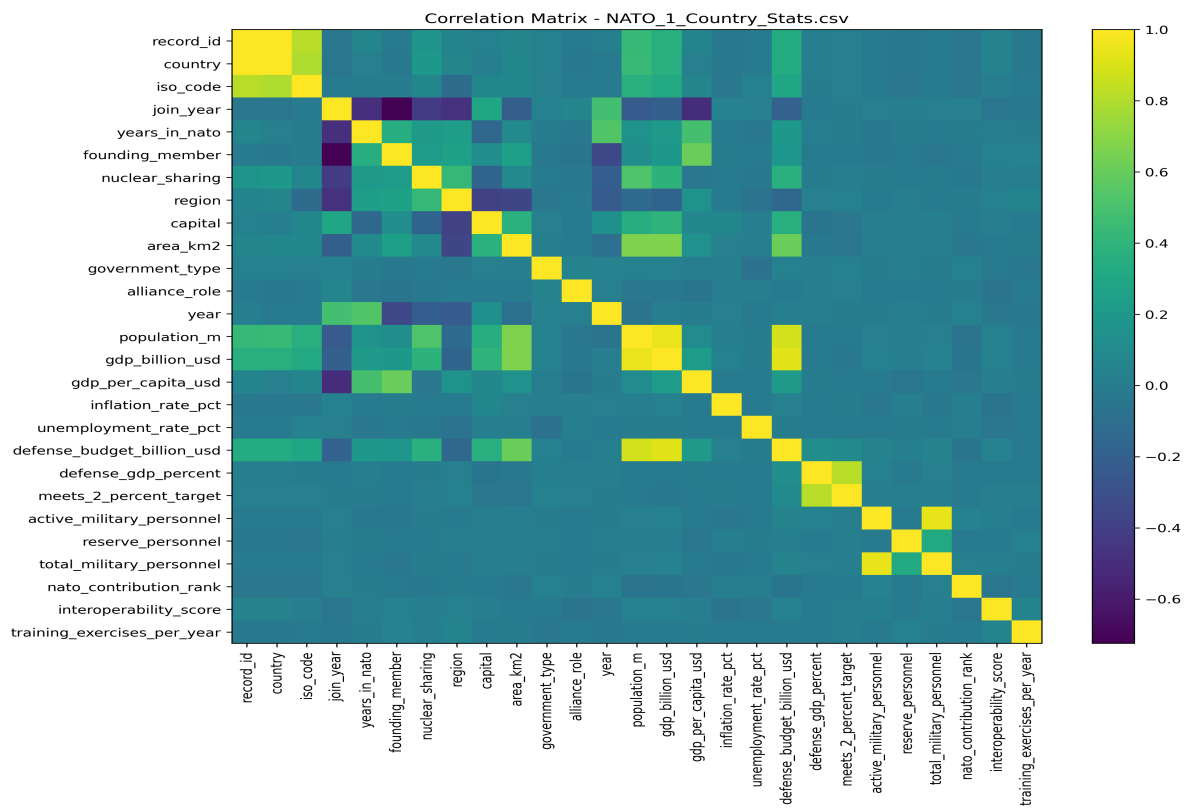
NATO Alliance Complete Dataset 2024

Comprehensive Data Science and Machine Learning Analysis

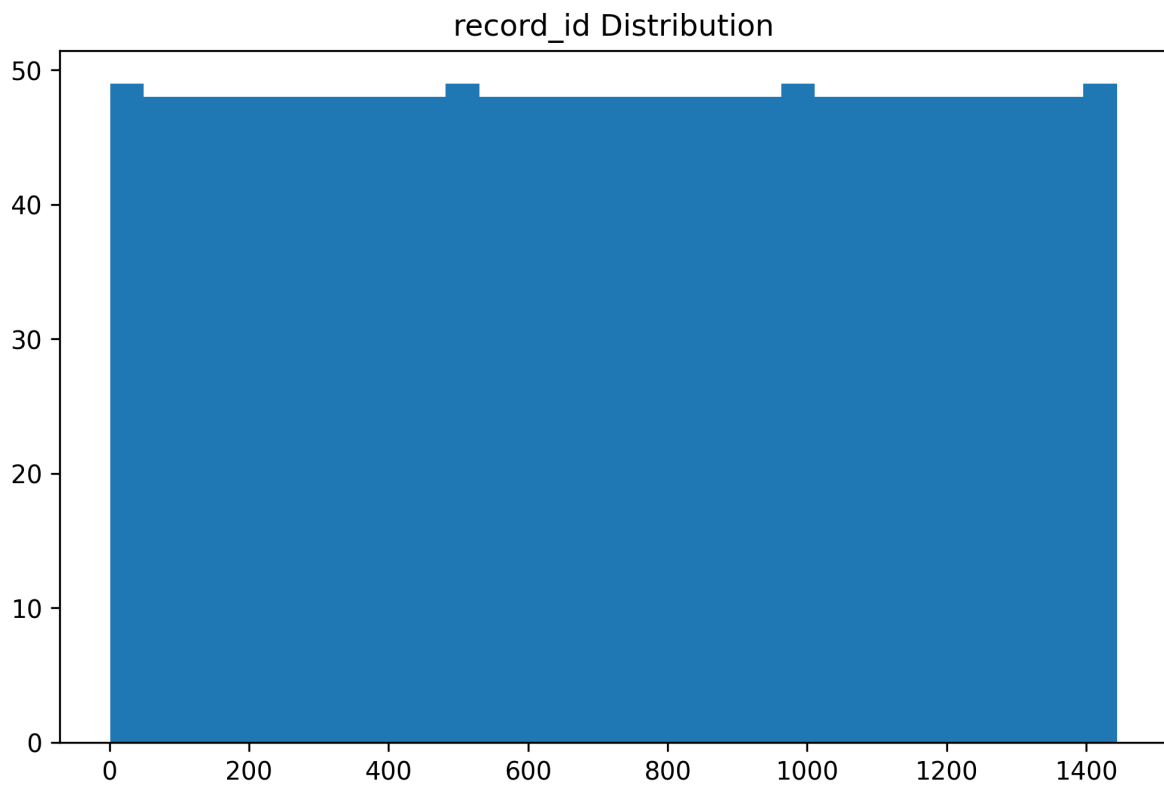
This report presents automated exploratory data analysis, machine learning insights, clustering analysis, anomaly detection, and statistical visualizations generated from the NATO Alliance Complete Dataset 2024.

Dataset Analysis: NATO_1_Country_Stats.csv

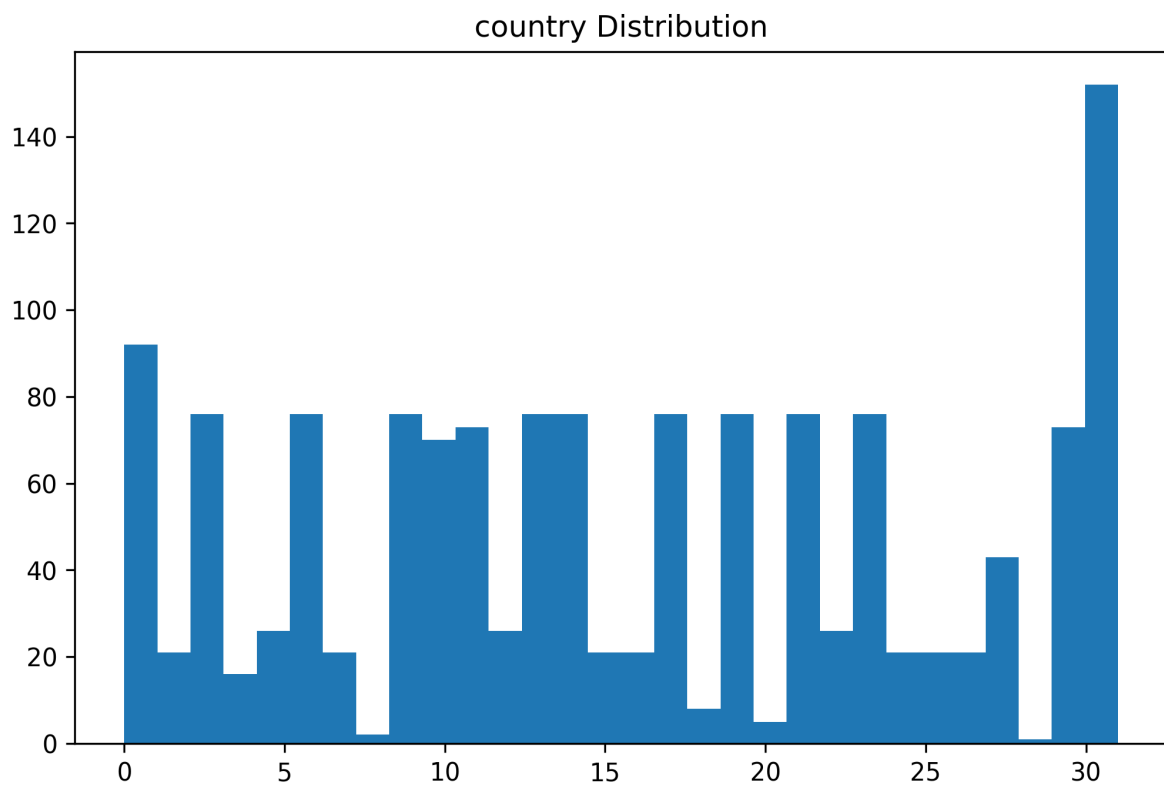
Rows: 1444
Columns: 29
Target Variable: gdp_billion_usd
Numerical Features: 27
Generated Charts: 21
Random Forest RMSE: 317.512
Gradient Boosting RMSE: 305.35



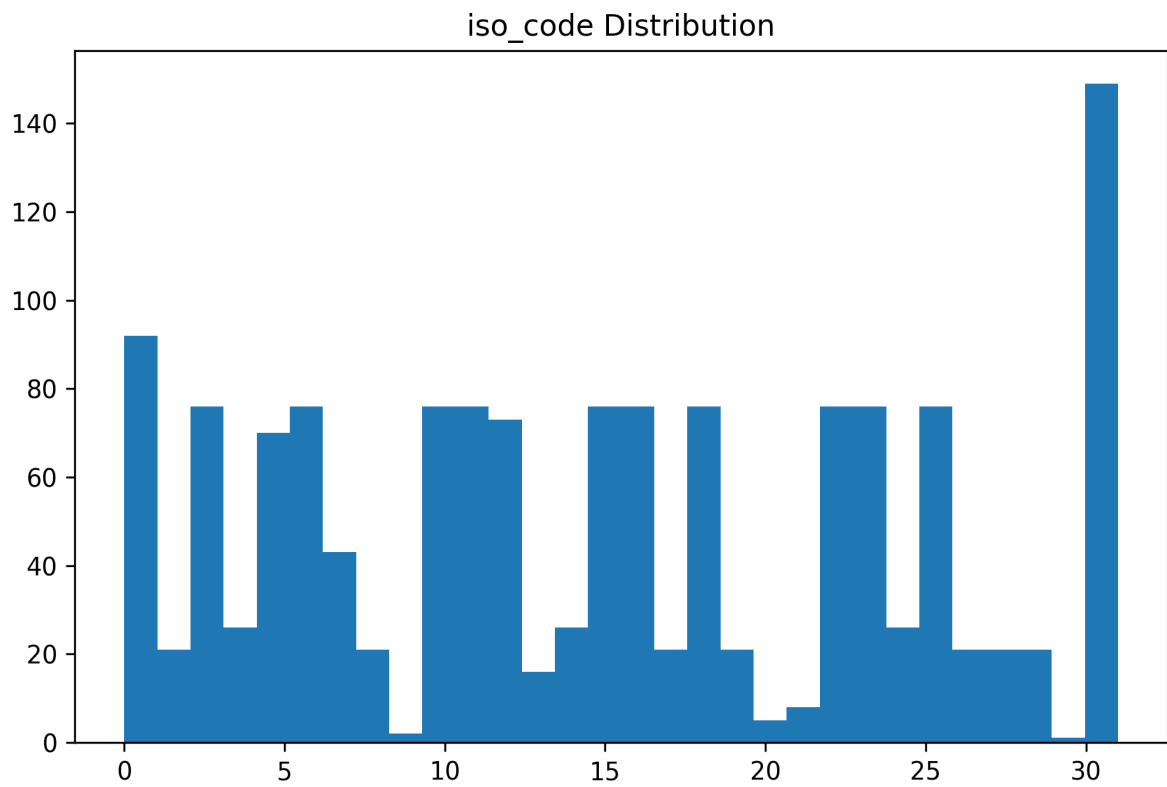
Correlation analysis between numerical variables showing positive and negative feature relationships.



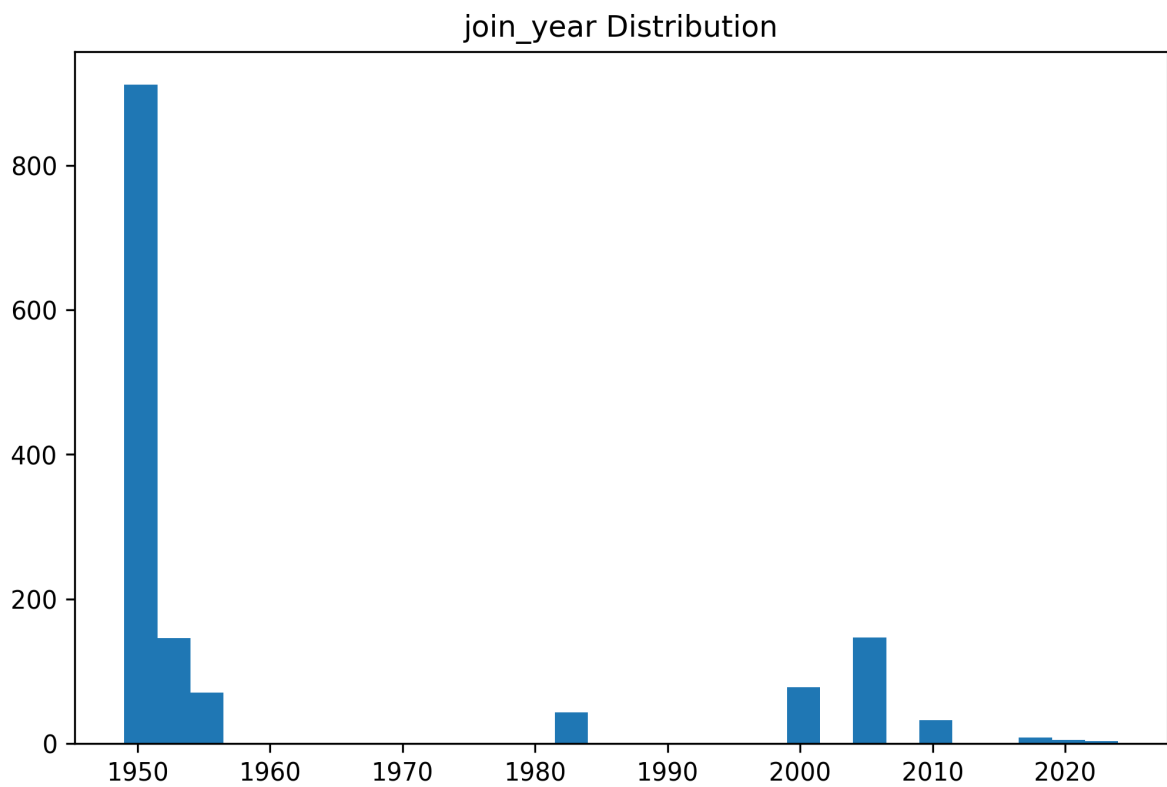
Distribution analysis for record_id showing value spread and density patterns.



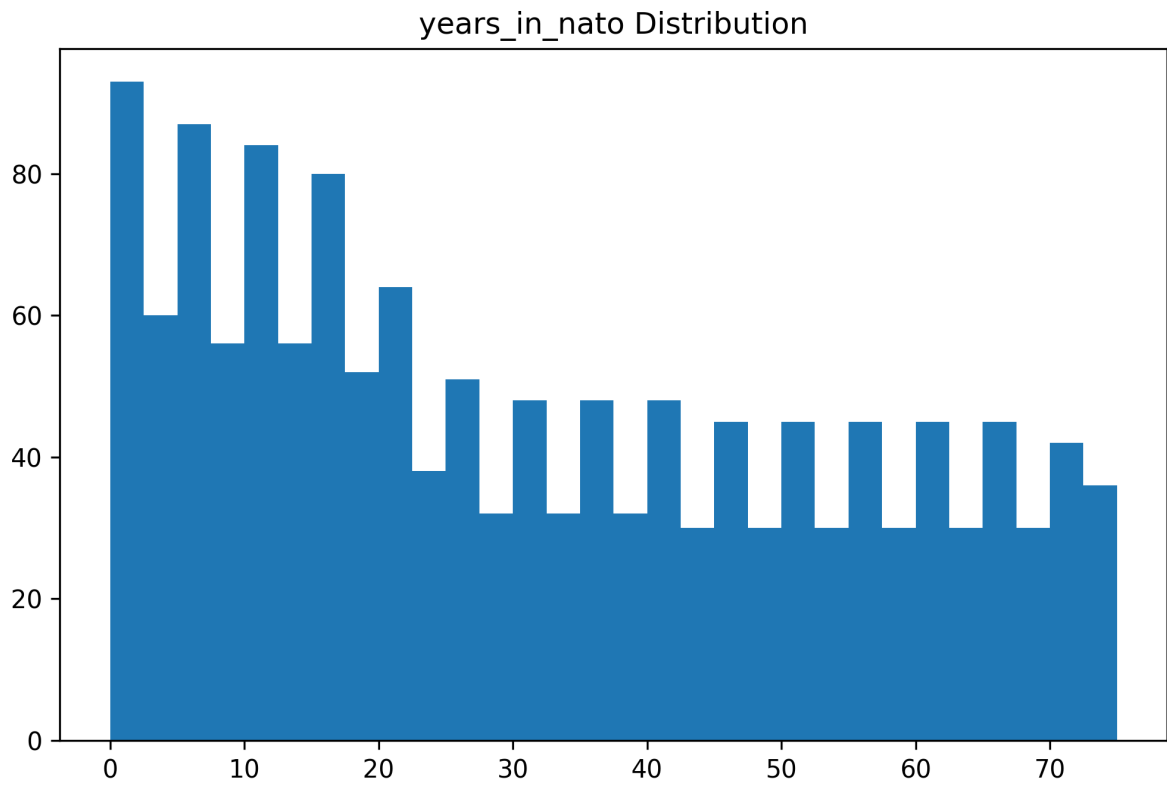
Distribution analysis for country showing value spread and density patterns.



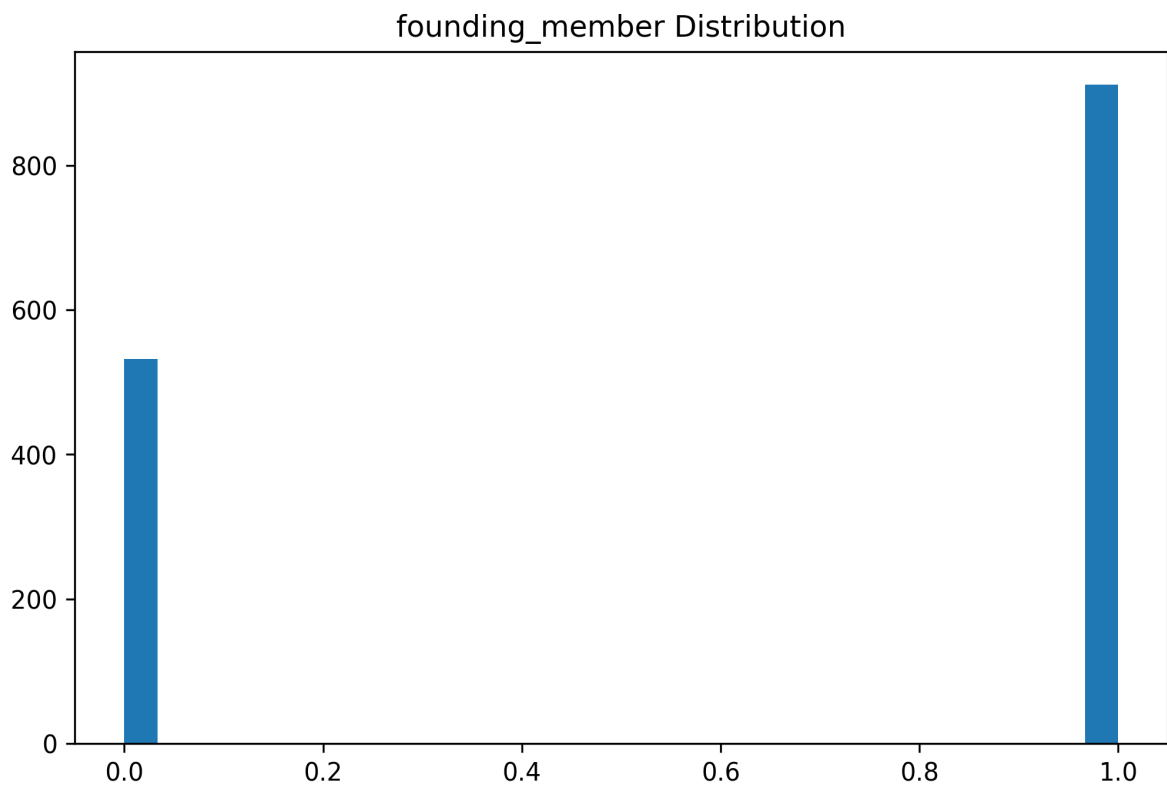
Distribution analysis for iso_code showing value spread and density patterns.



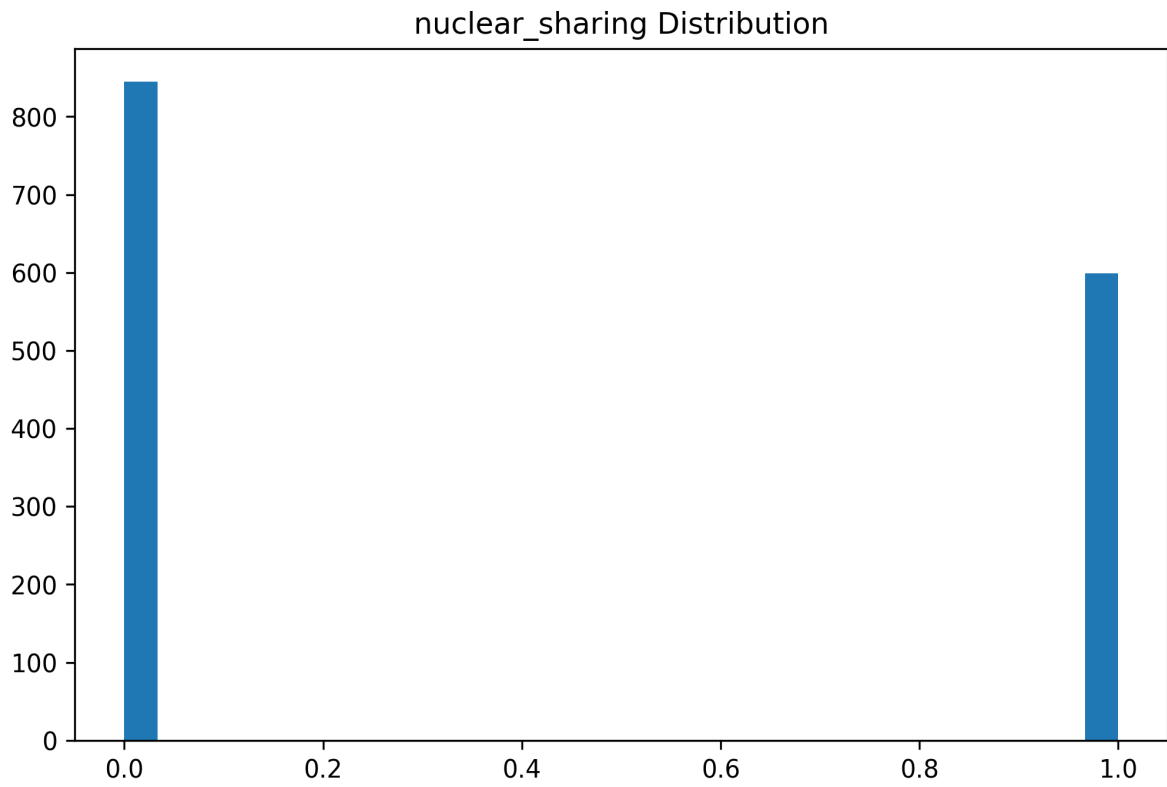
Distribution analysis for join_year showing value spread and density patterns.



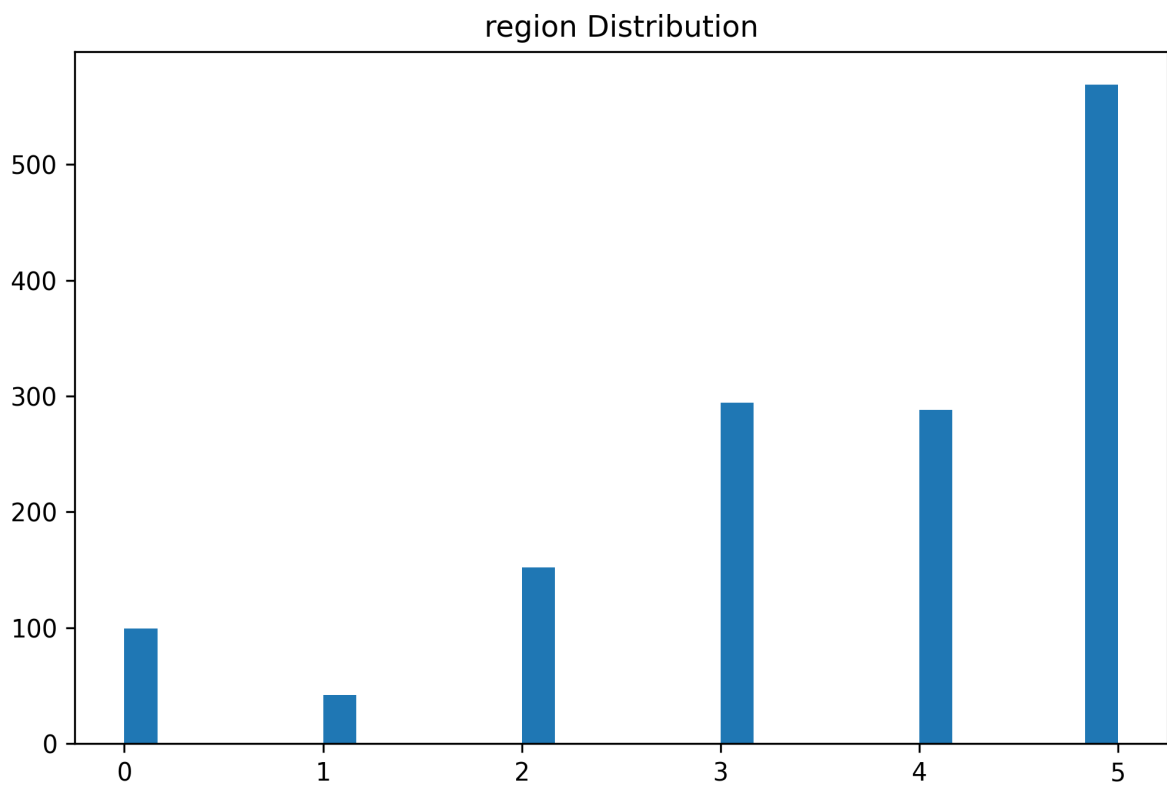
Distribution analysis for years_in_nato showing value spread and density patterns.



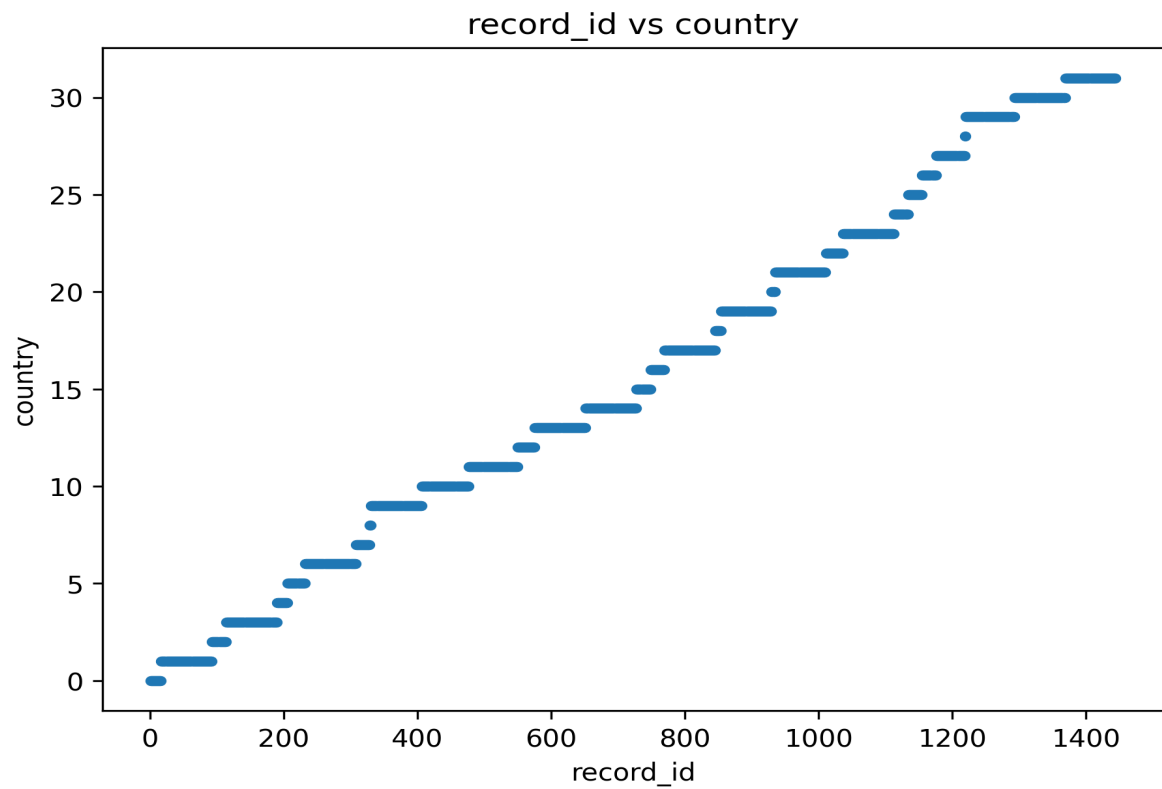
Distribution analysis for founding_member showing value spread and density patterns.



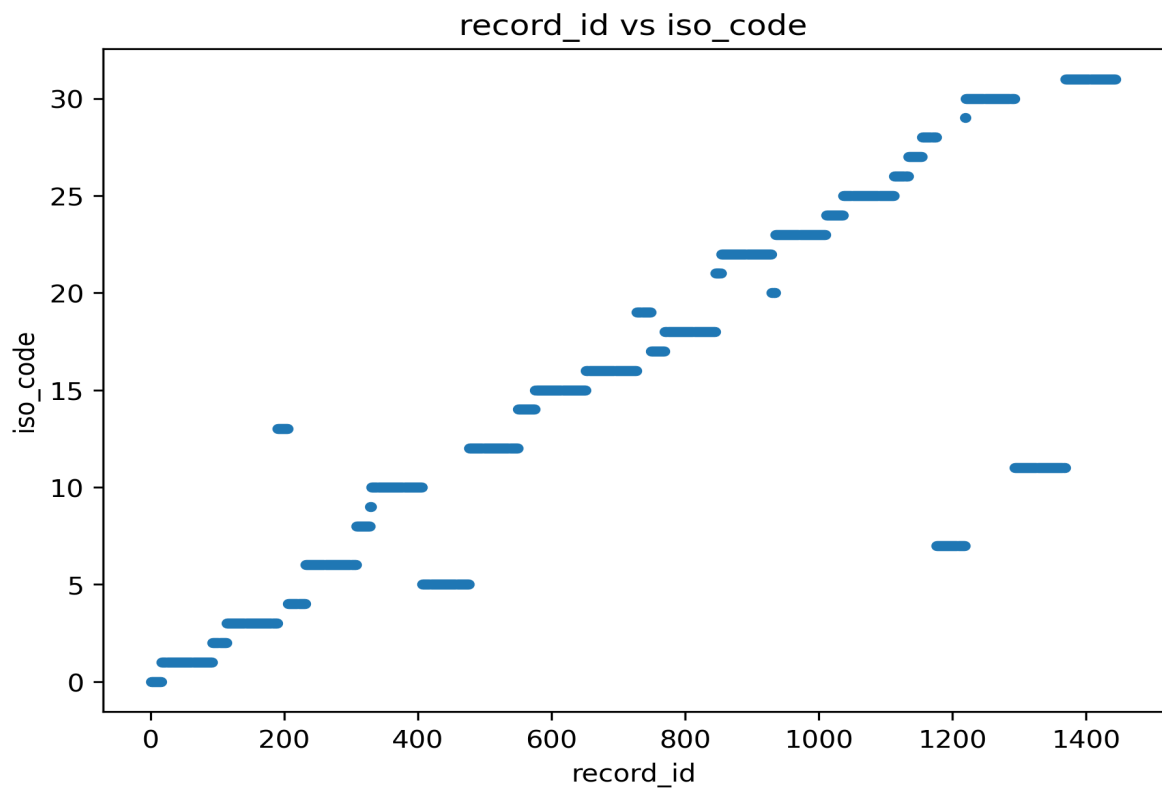
Distribution analysis for nuclear_sharing showing value spread and density patterns.



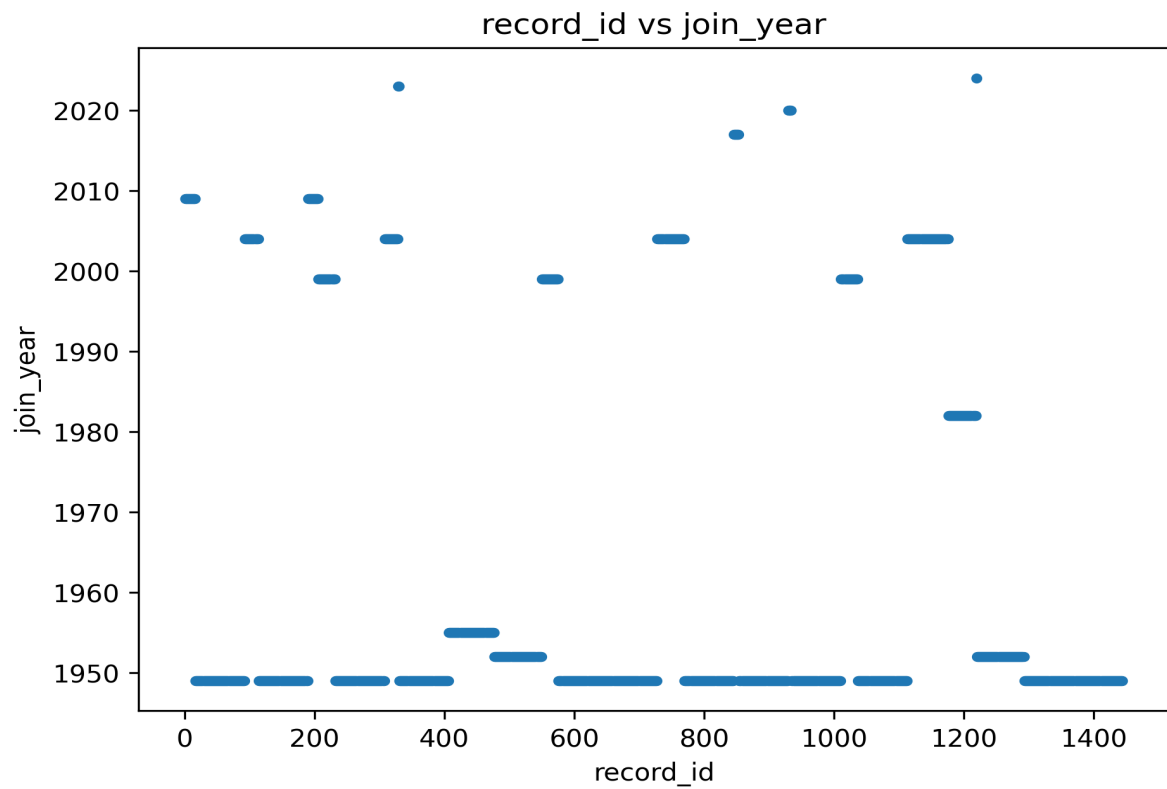
Distribution analysis for region showing value spread and density patterns.



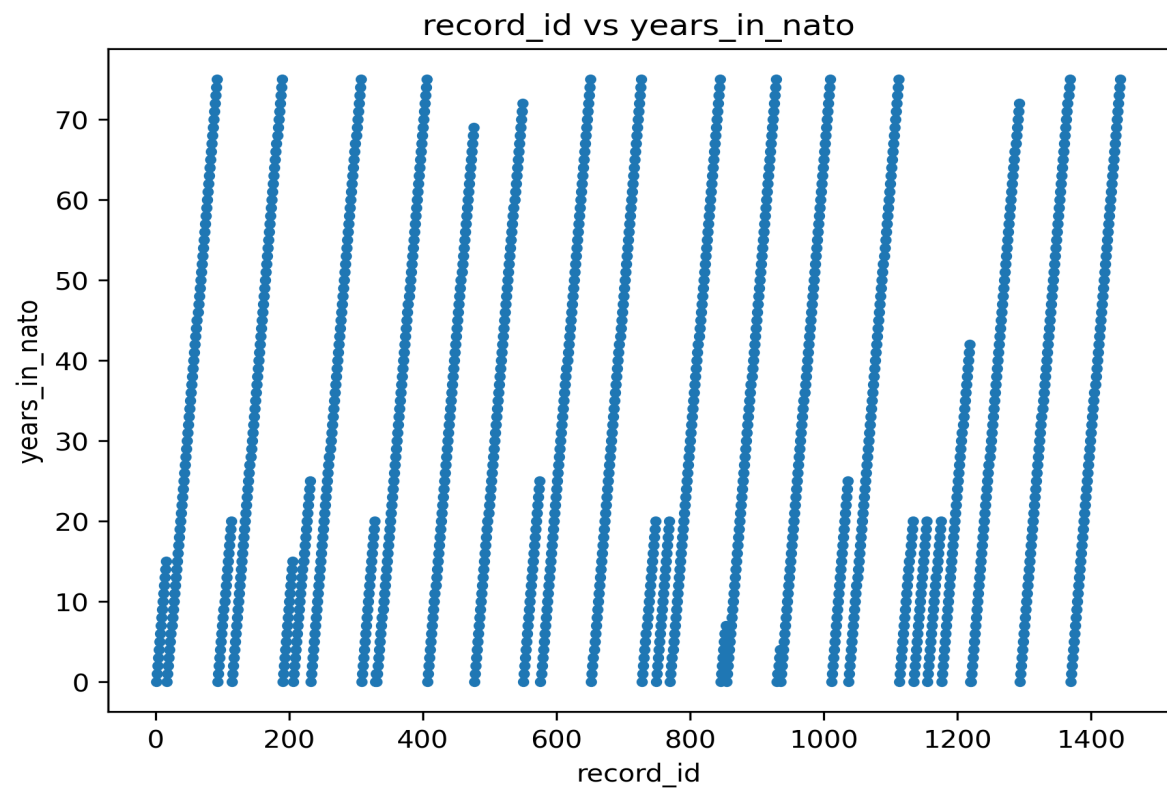
Scatter relationship between record_id and country.



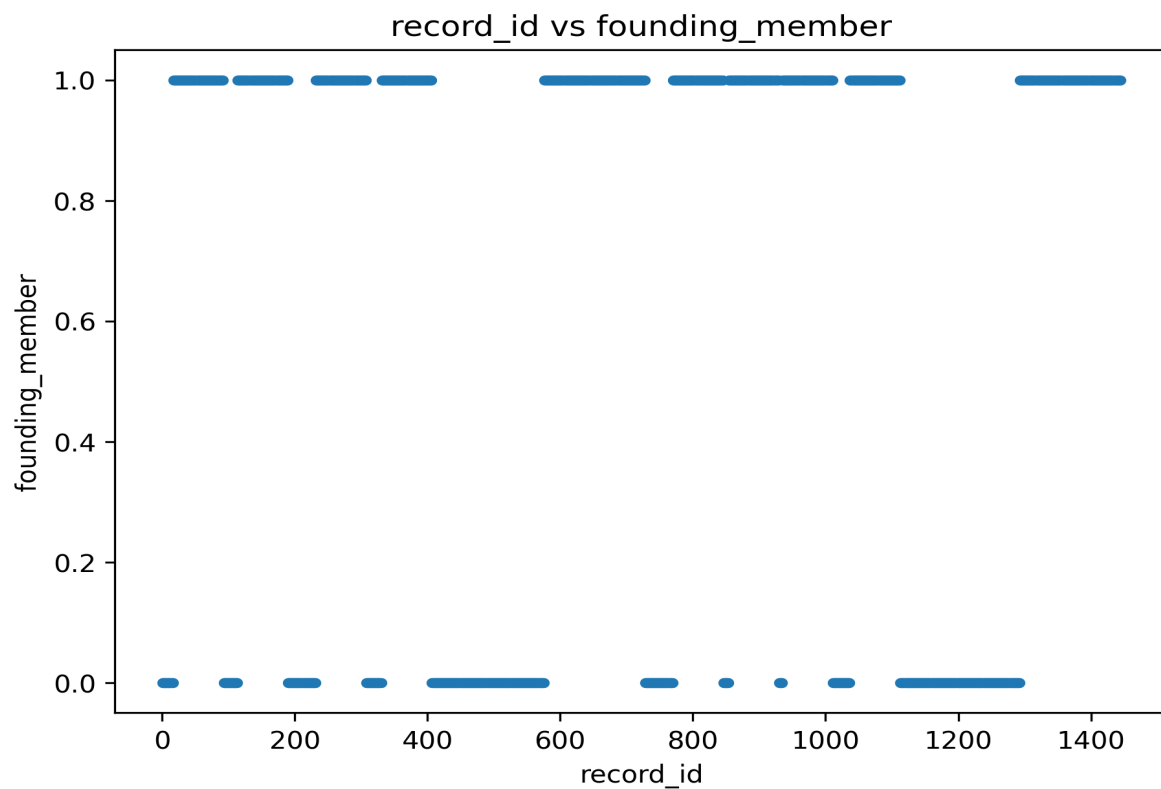
Scatter relationship between record_id and iso_code.



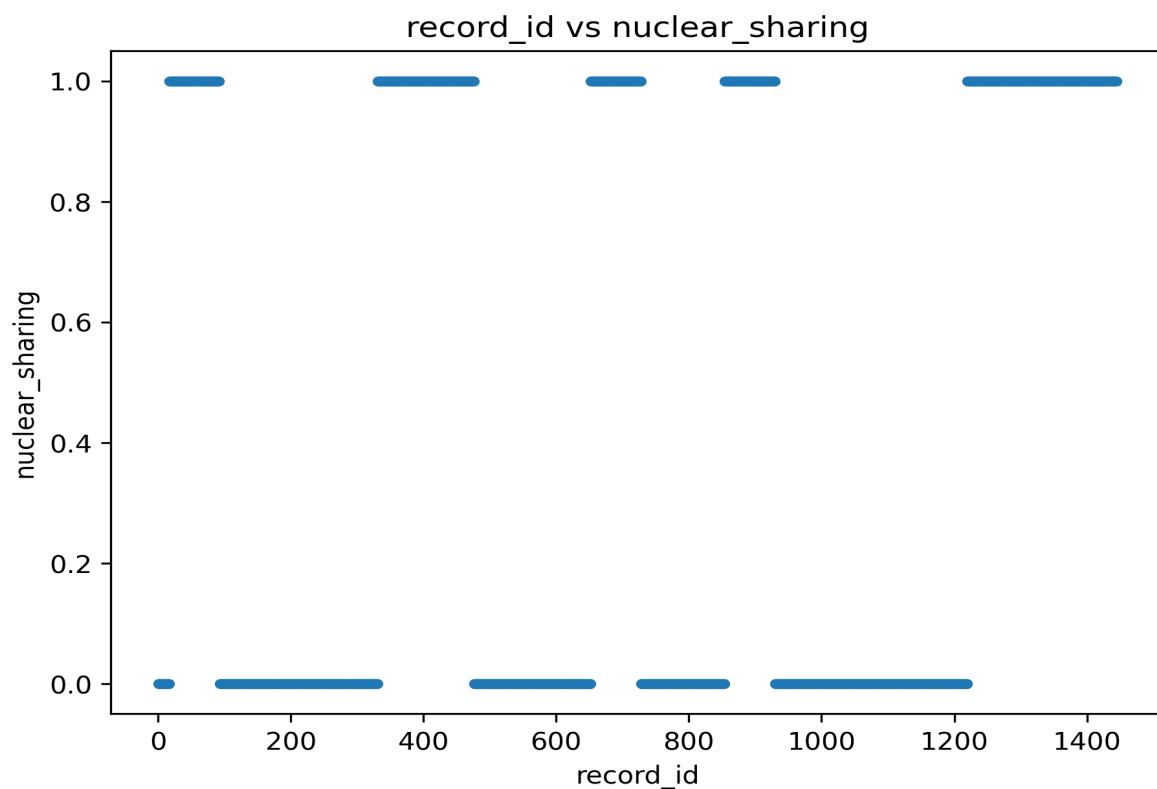
Scatter relationship between record_id and join_year.



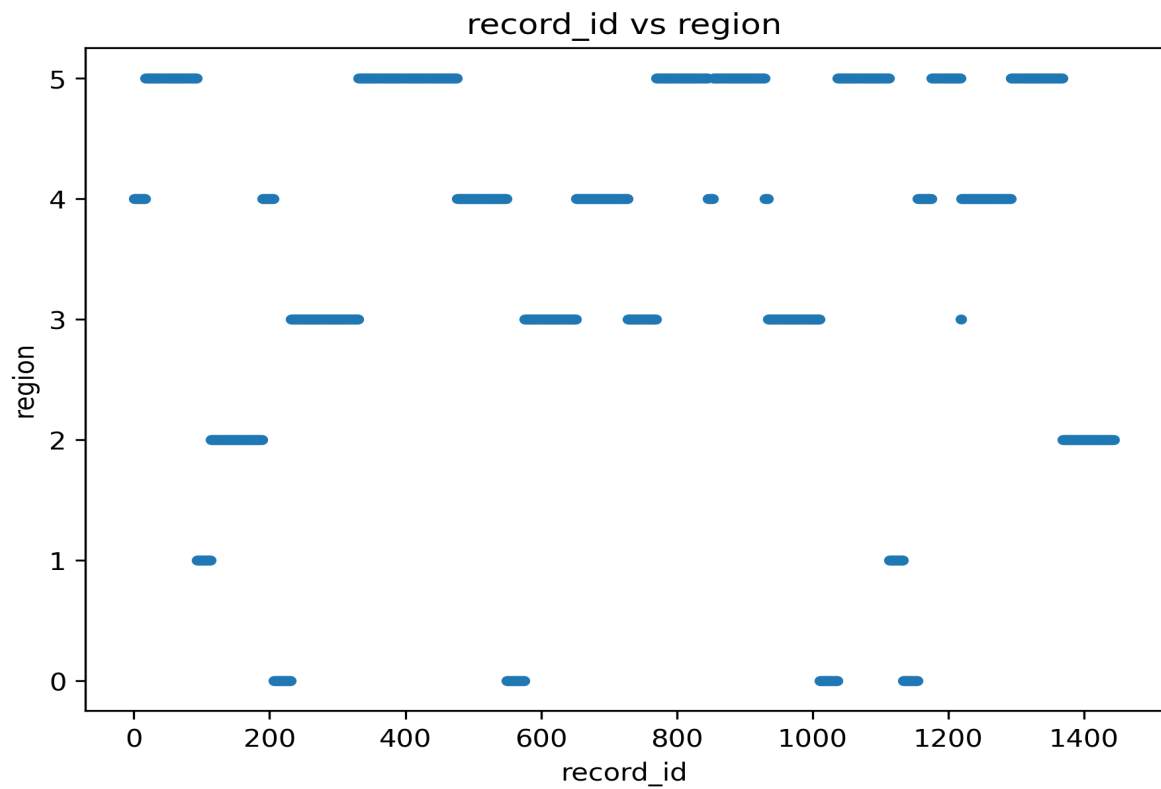
Scatter relationship between record_id and years_in_nato.



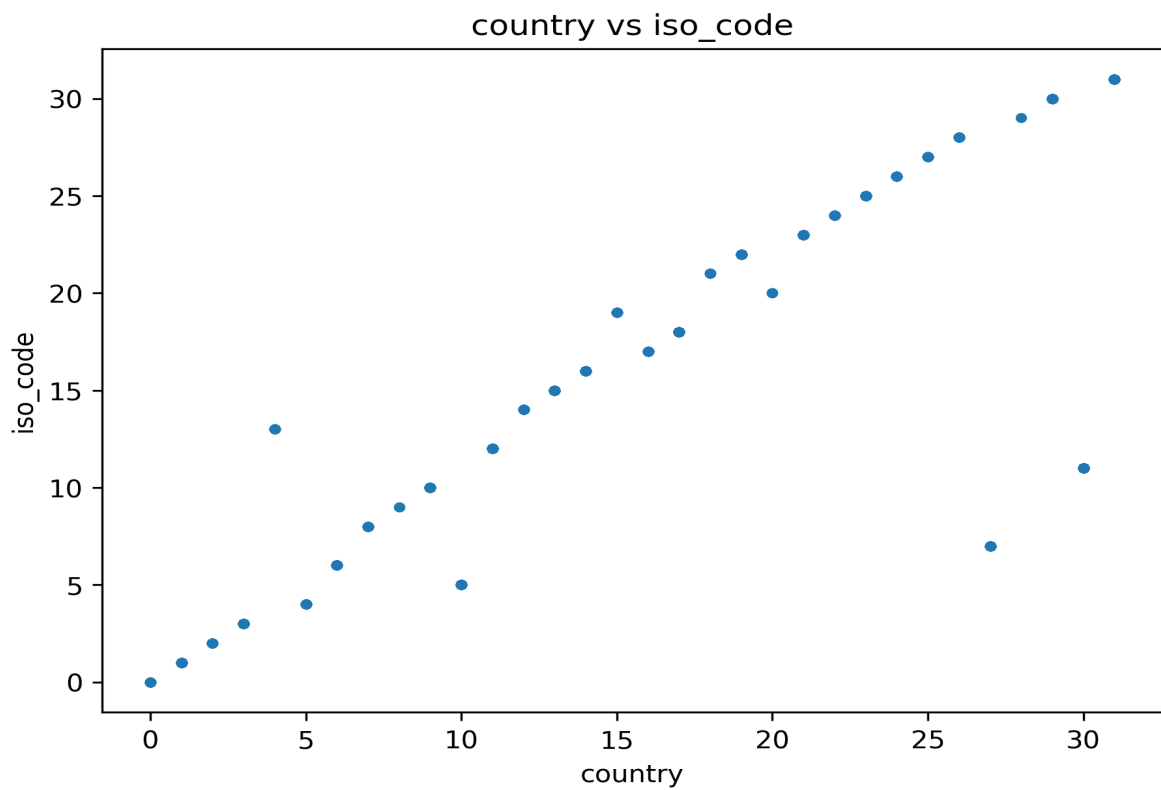
Scatter relationship between record_id and founding_member.



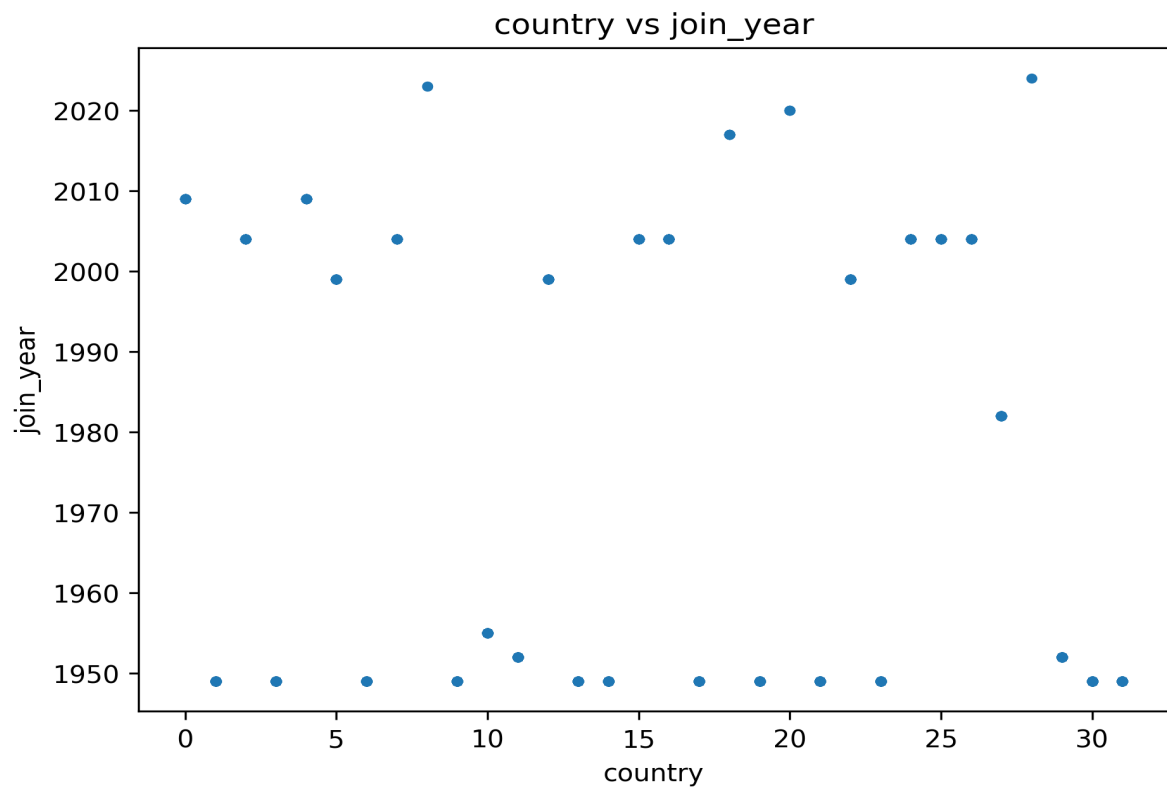
Scatter relationship between record_id and nuclear_sharing.

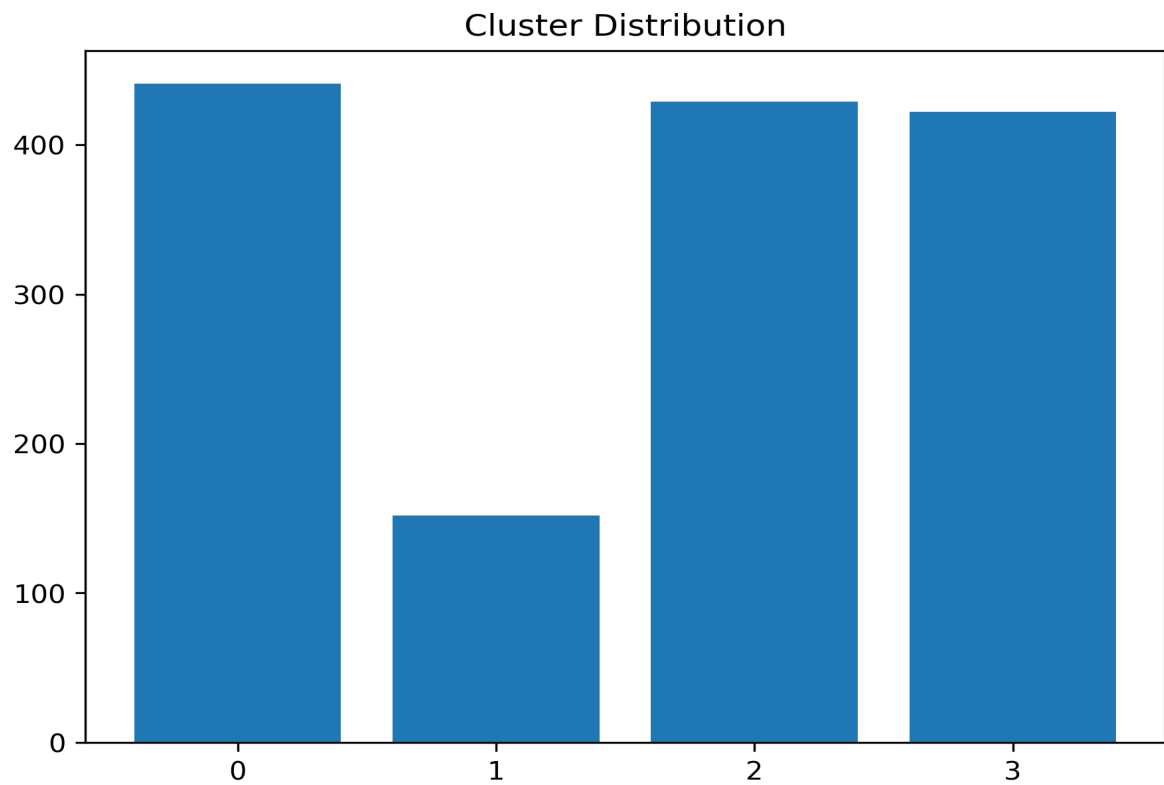


Scatter relationship between record_id and region.

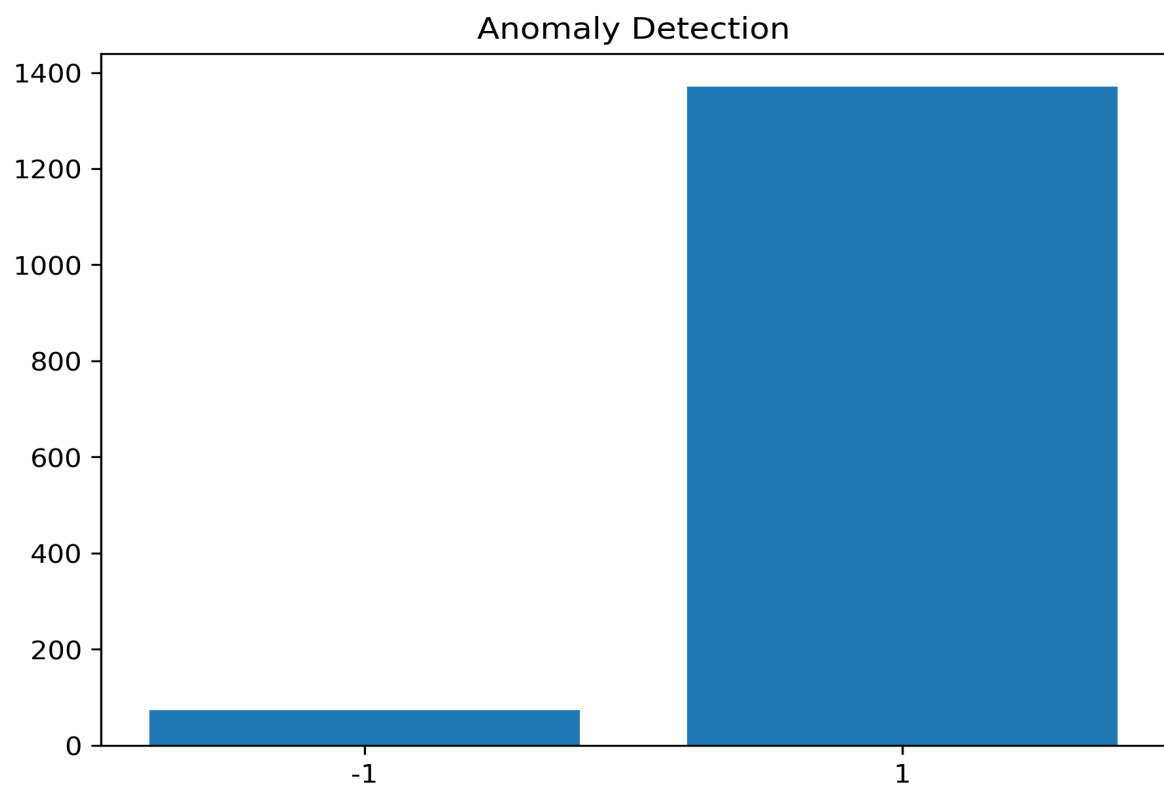


Scatter relationship between country and iso_code.





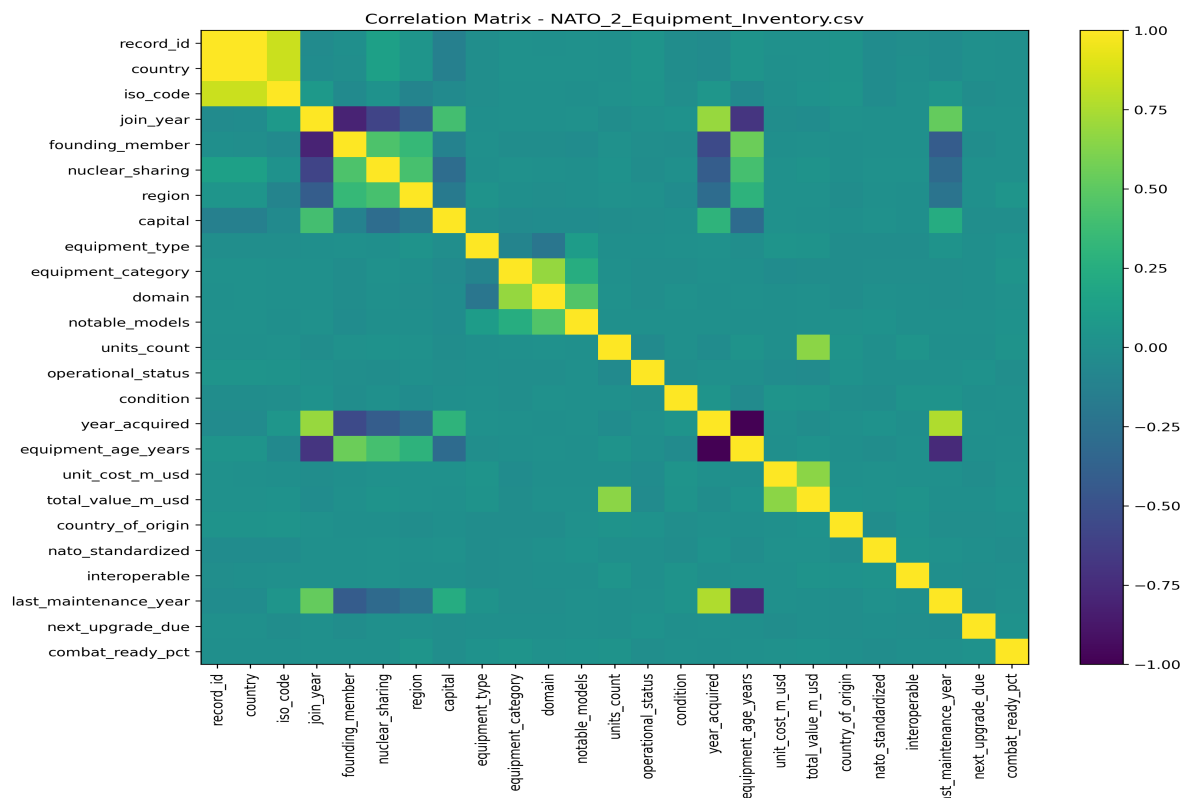
KMeans clustering groups similar records into data-driven clusters.



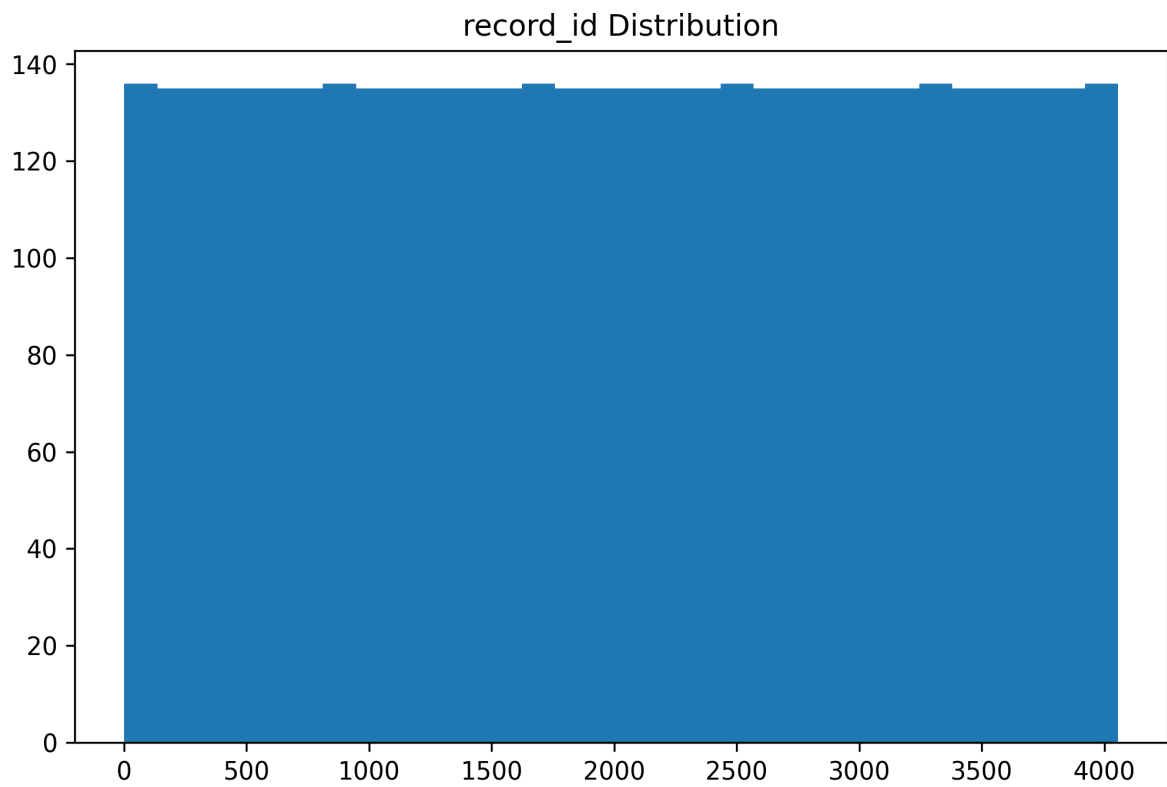
Isolation Forest identifies unusual or abnormal observations within the dataset.

Dataset Analysis: NATO_2_Equipment_Inventory.csv

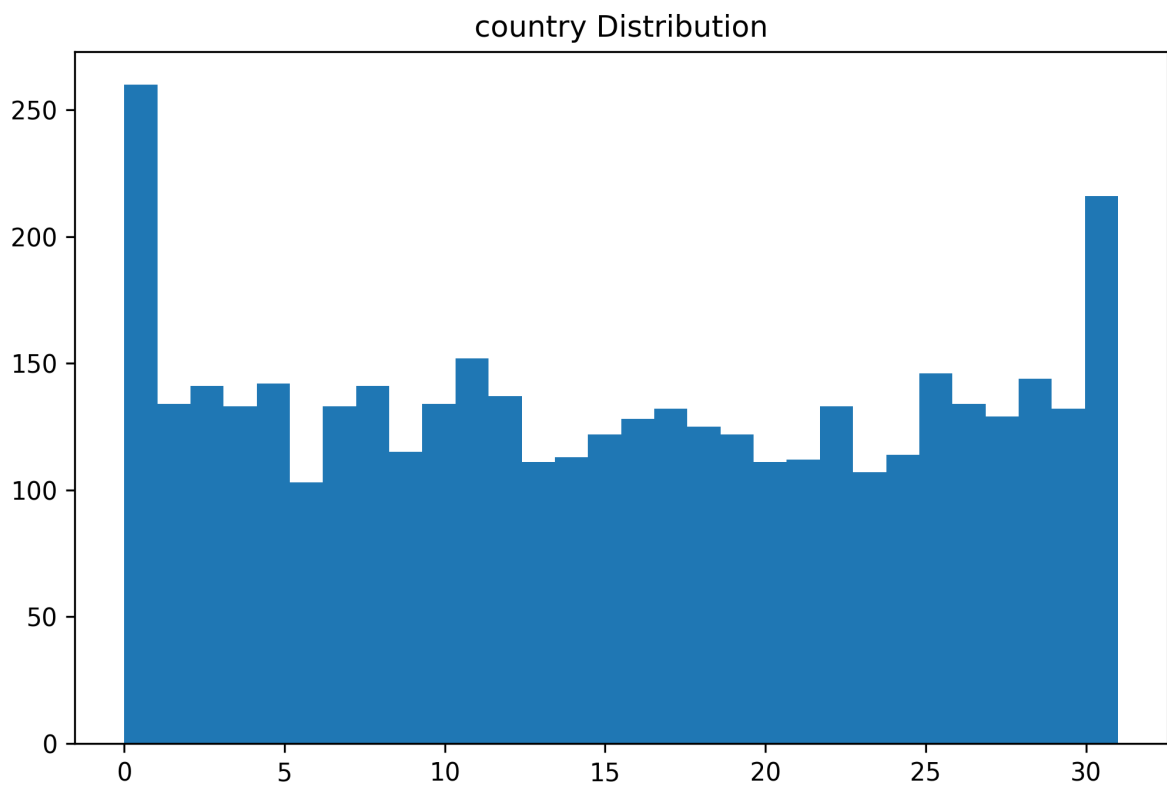
Rows: 4056
Columns: 27
Target Variable: units_count
Numerical Features: 25
Generated Charts: 21
Random Forest RMSE: 45.657
Gradient Boosting RMSE: 82.246



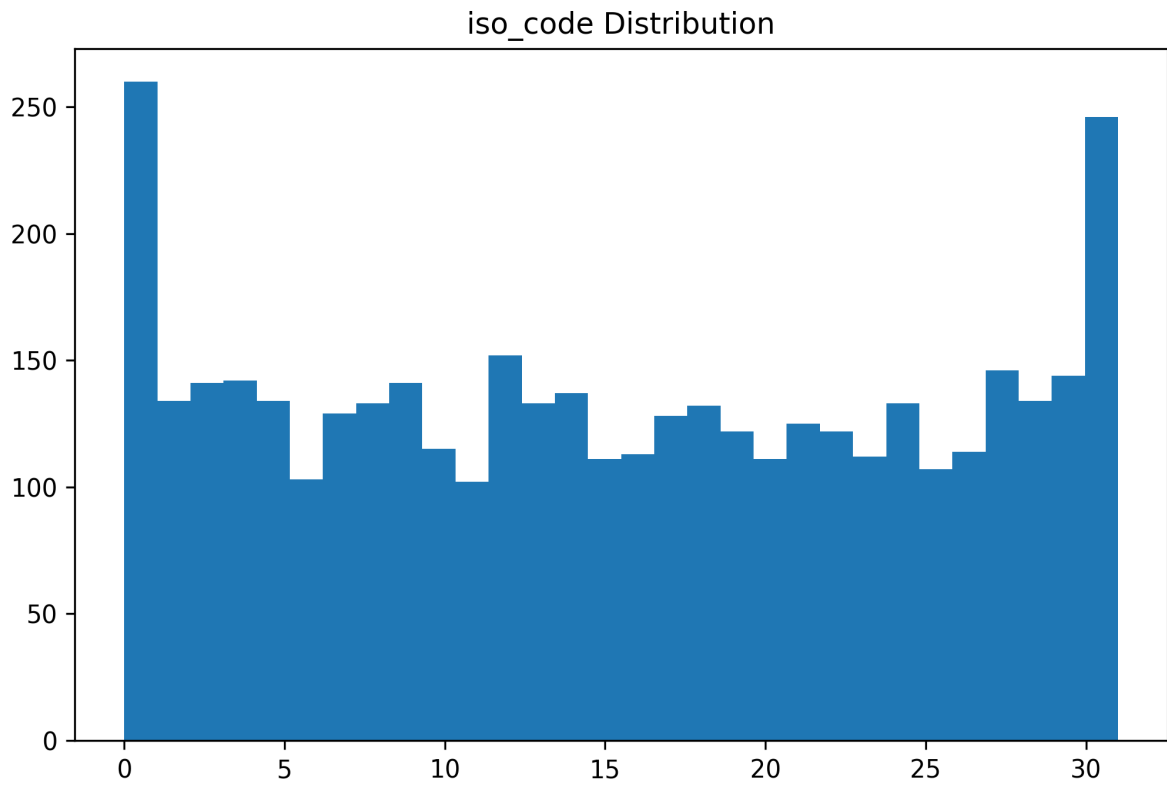
Correlation analysis between numerical variables showing positive and negative feature relationships.



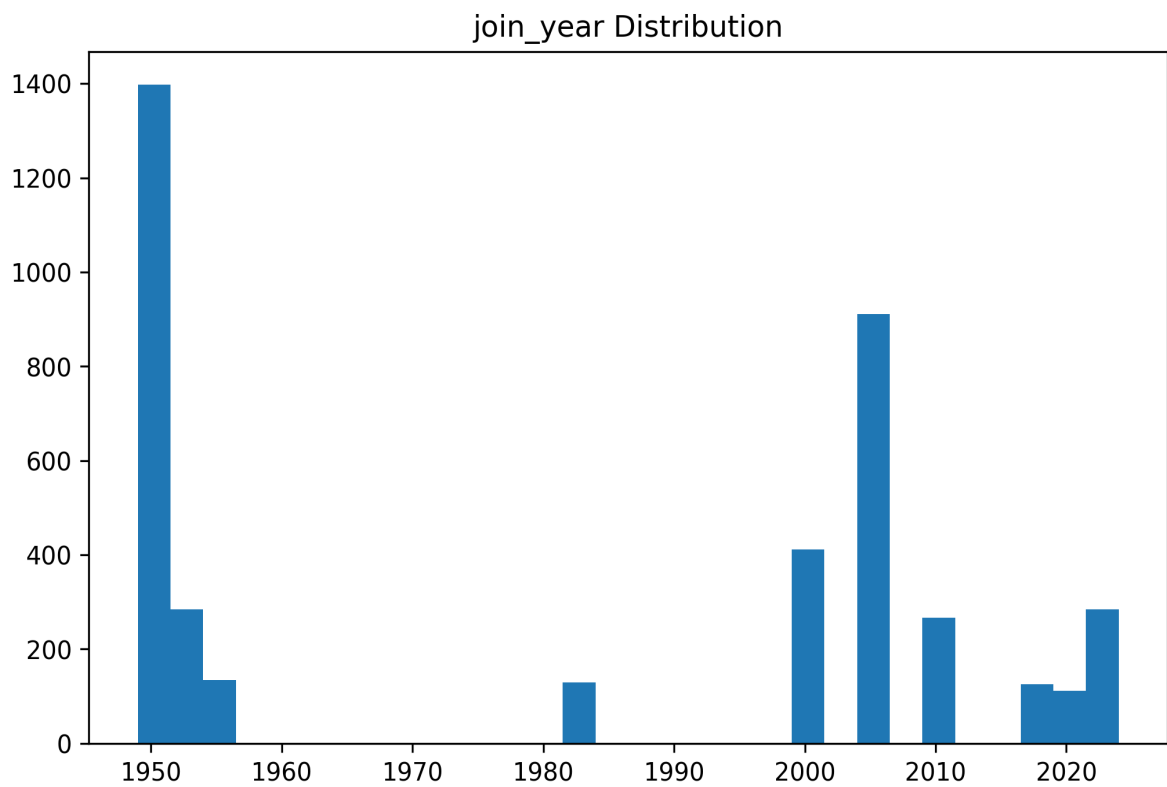
Distribution analysis for record_id showing value spread and density patterns.



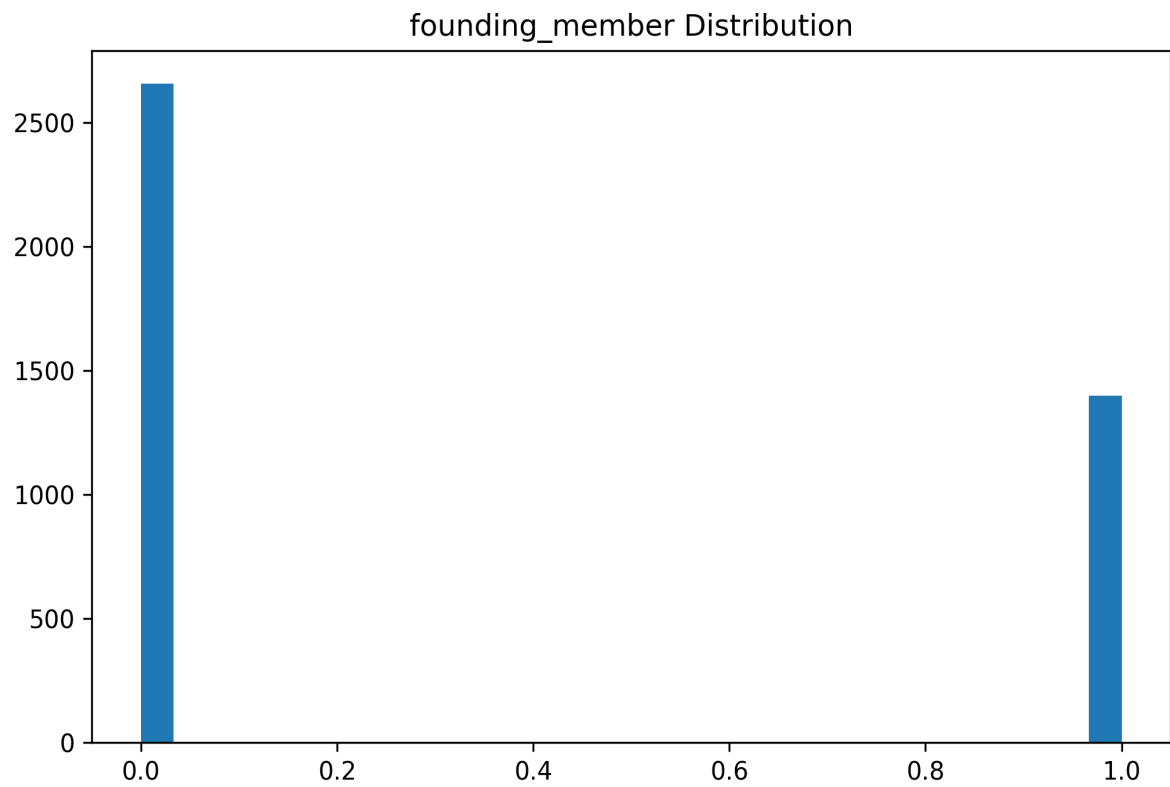
Distribution analysis for country showing value spread and density patterns.



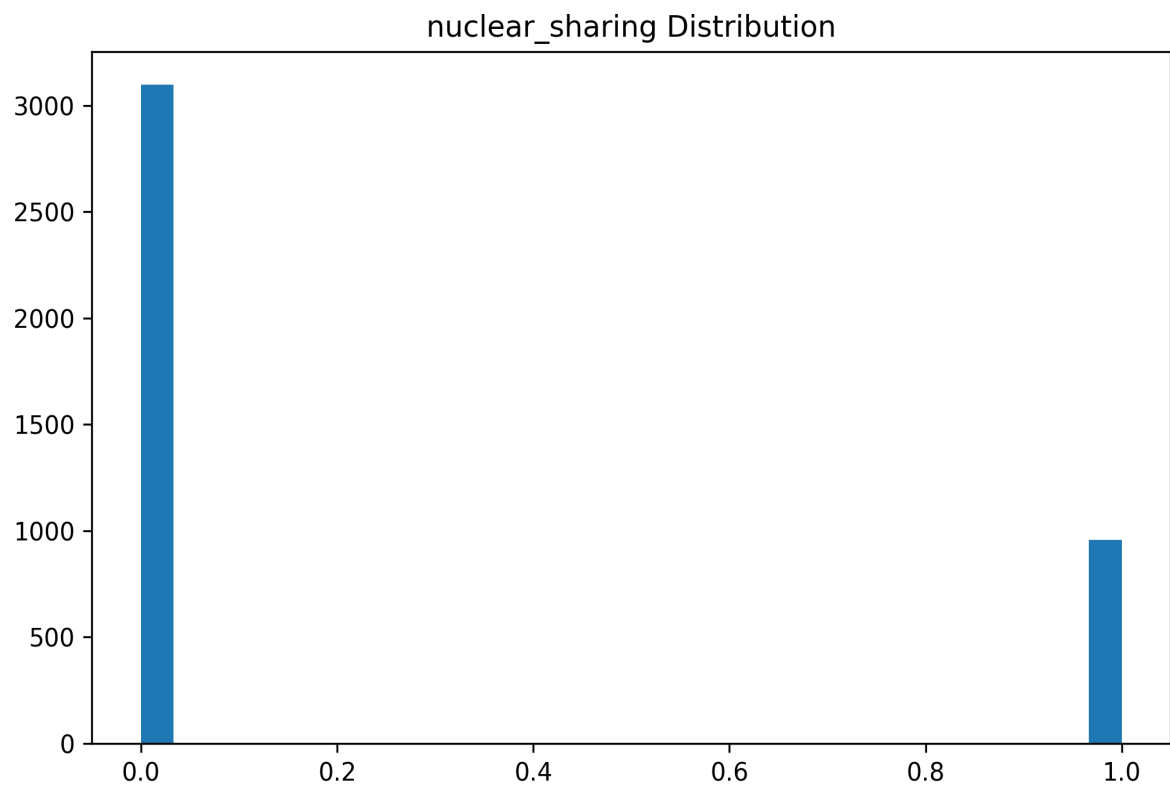
Distribution analysis for iso_code showing value spread and density patterns.



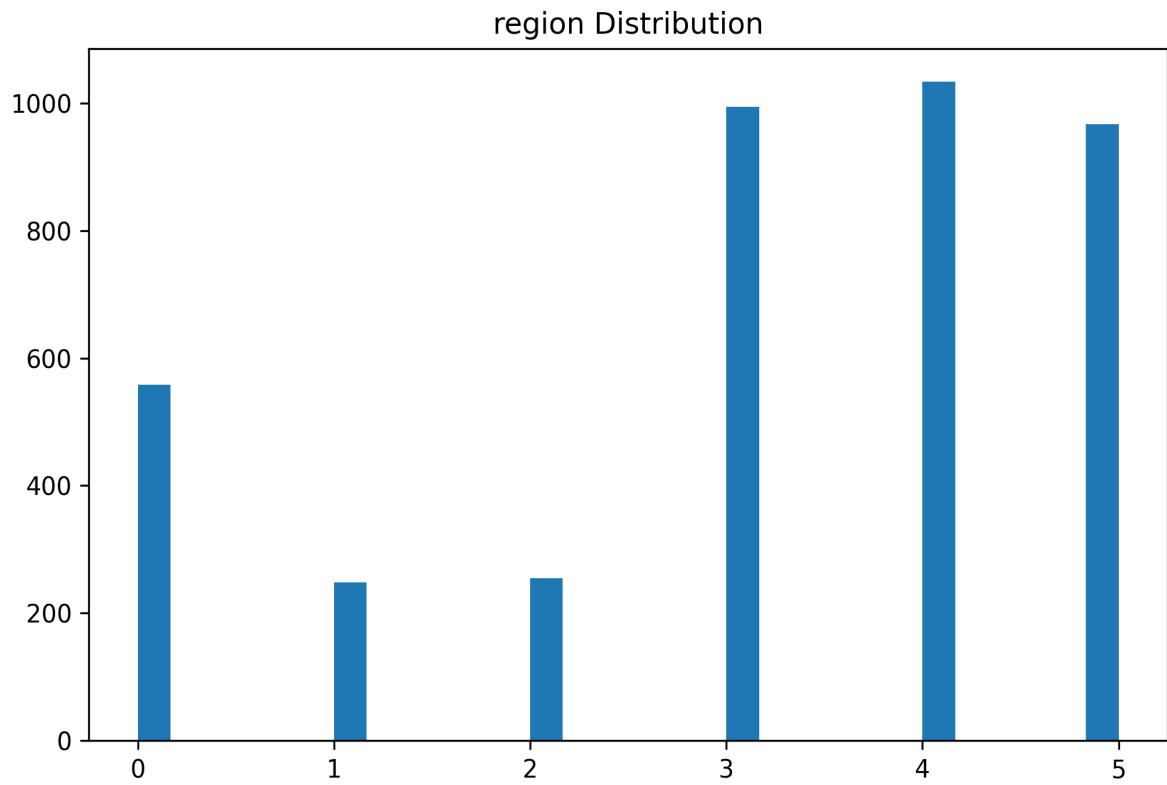
Distribution analysis for join_year showing value spread and density patterns.



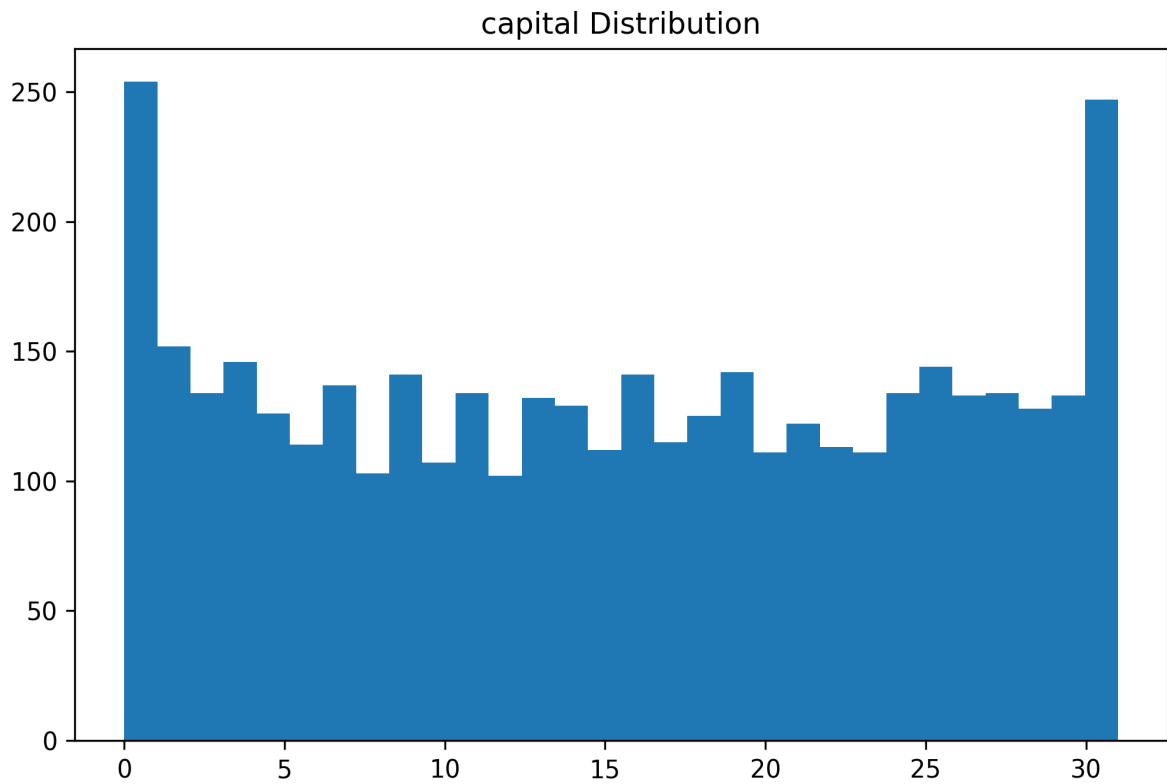
Distribution analysis for founding_member showing value spread and density patterns.



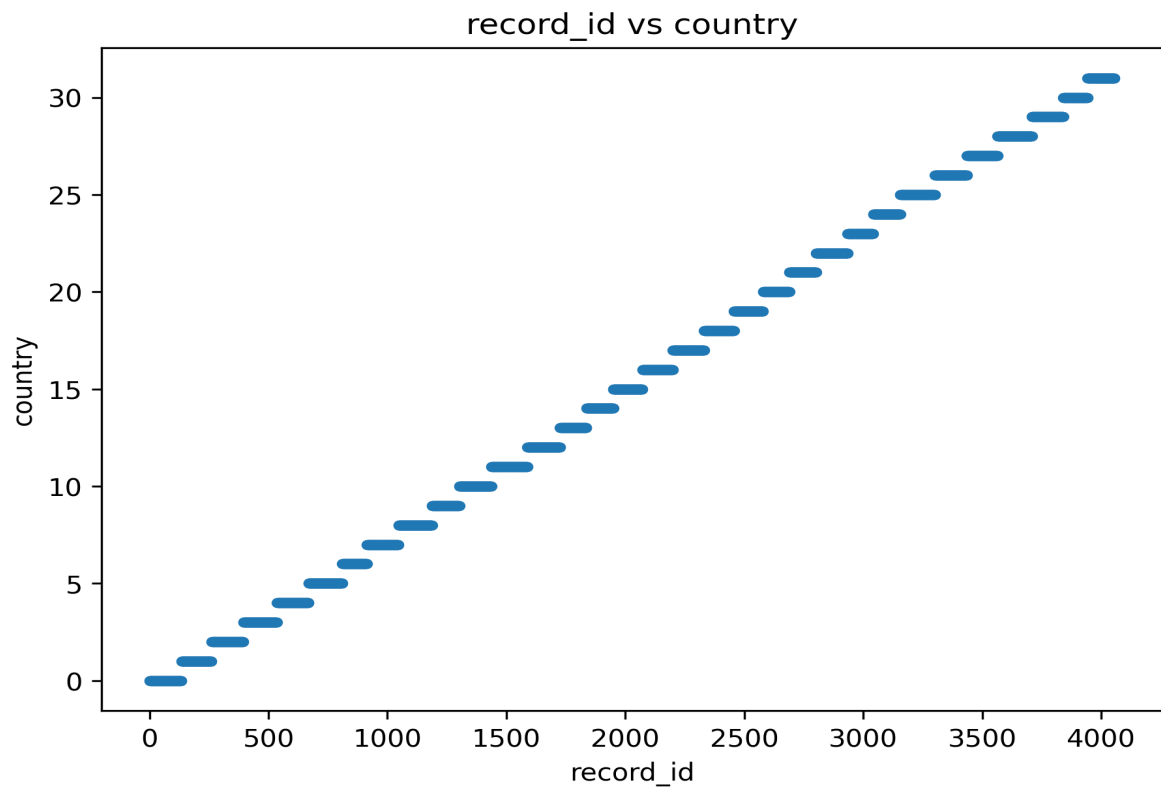
Distribution analysis for nuclear_sharing showing value spread and density patterns.



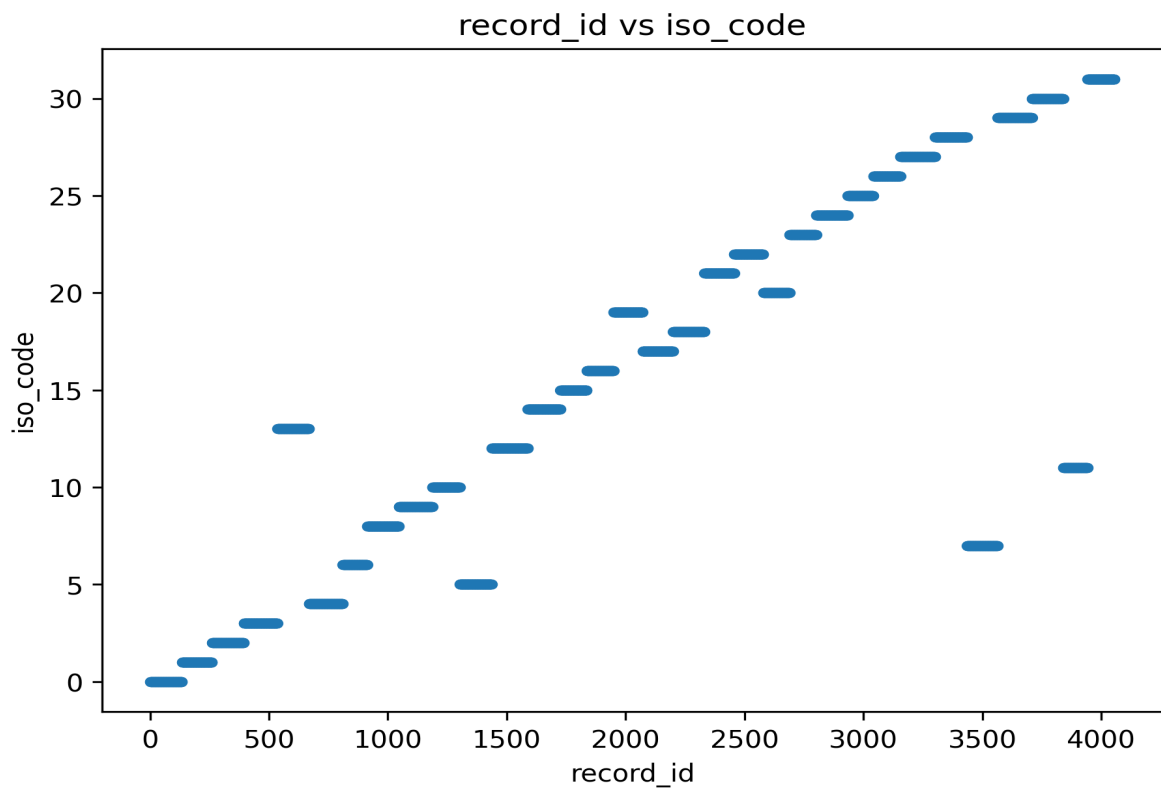
Distribution analysis for region showing value spread and density patterns.



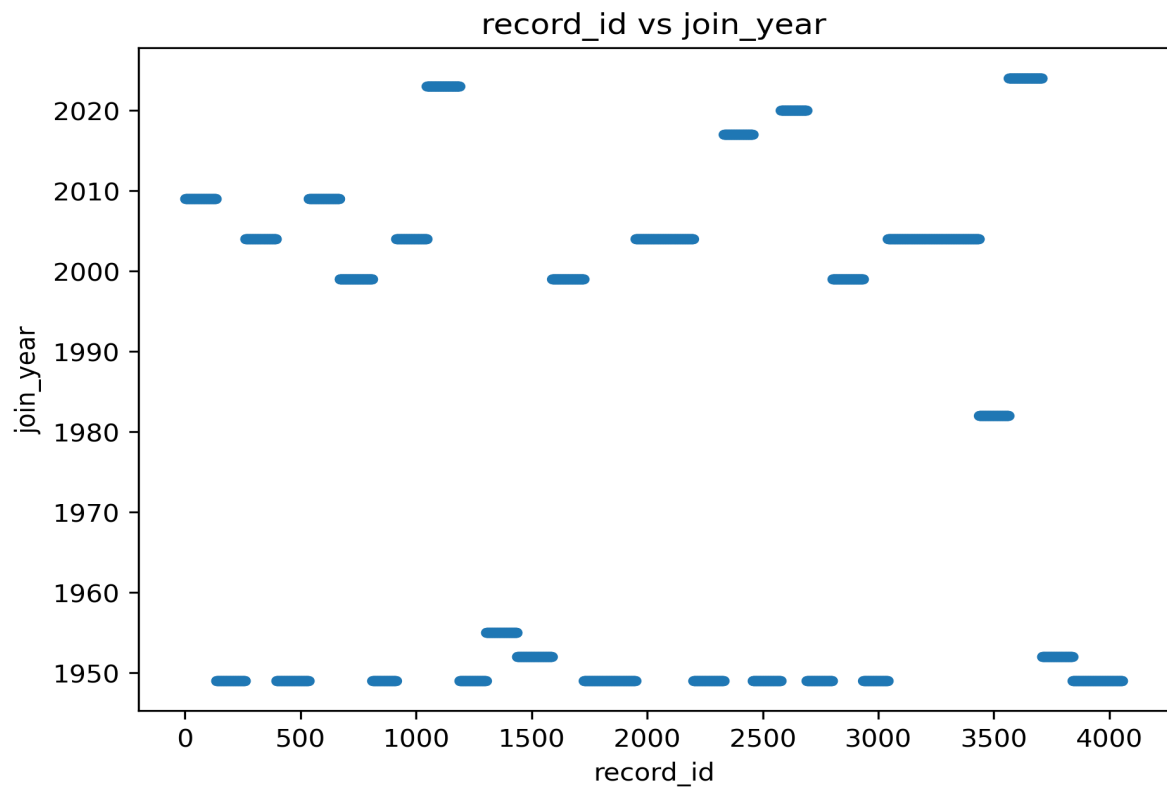
Distribution analysis for capital showing value spread and density patterns.



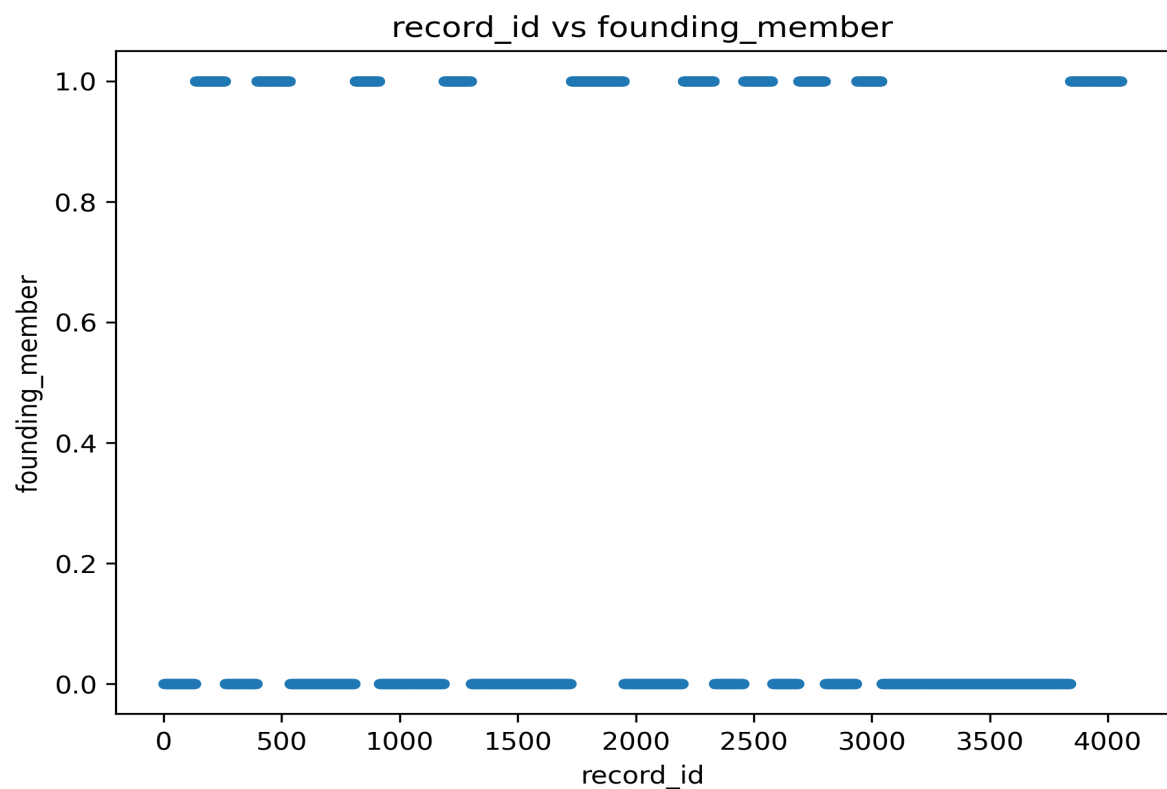
Scatter relationship between record_id and country.



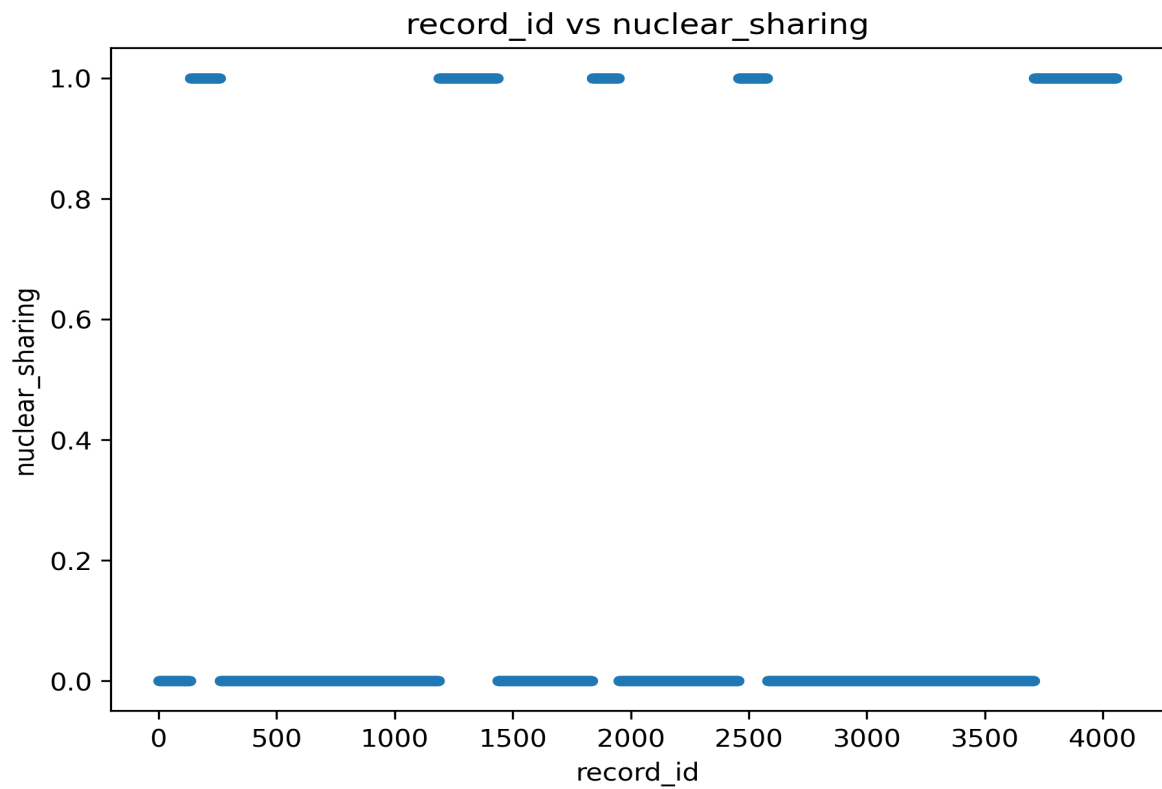
Scatter relationship between record_id and iso_code.



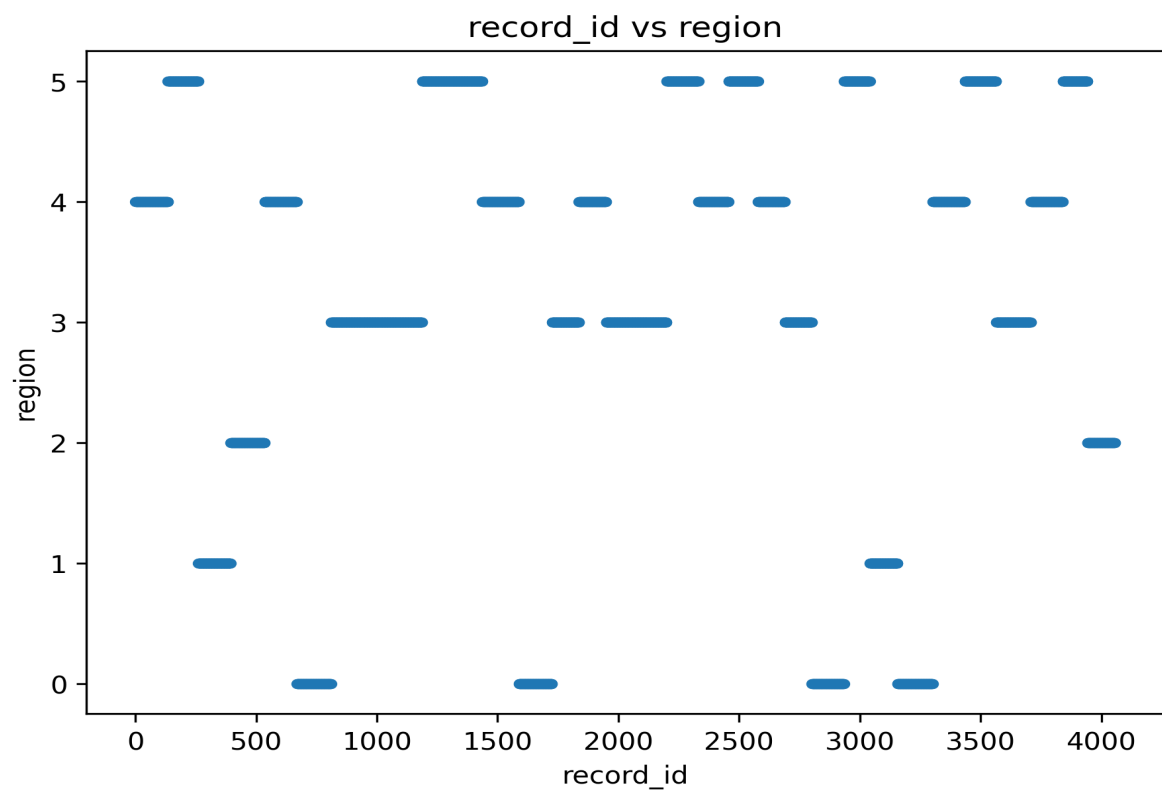
Scatter relationship between record_id and join_year.



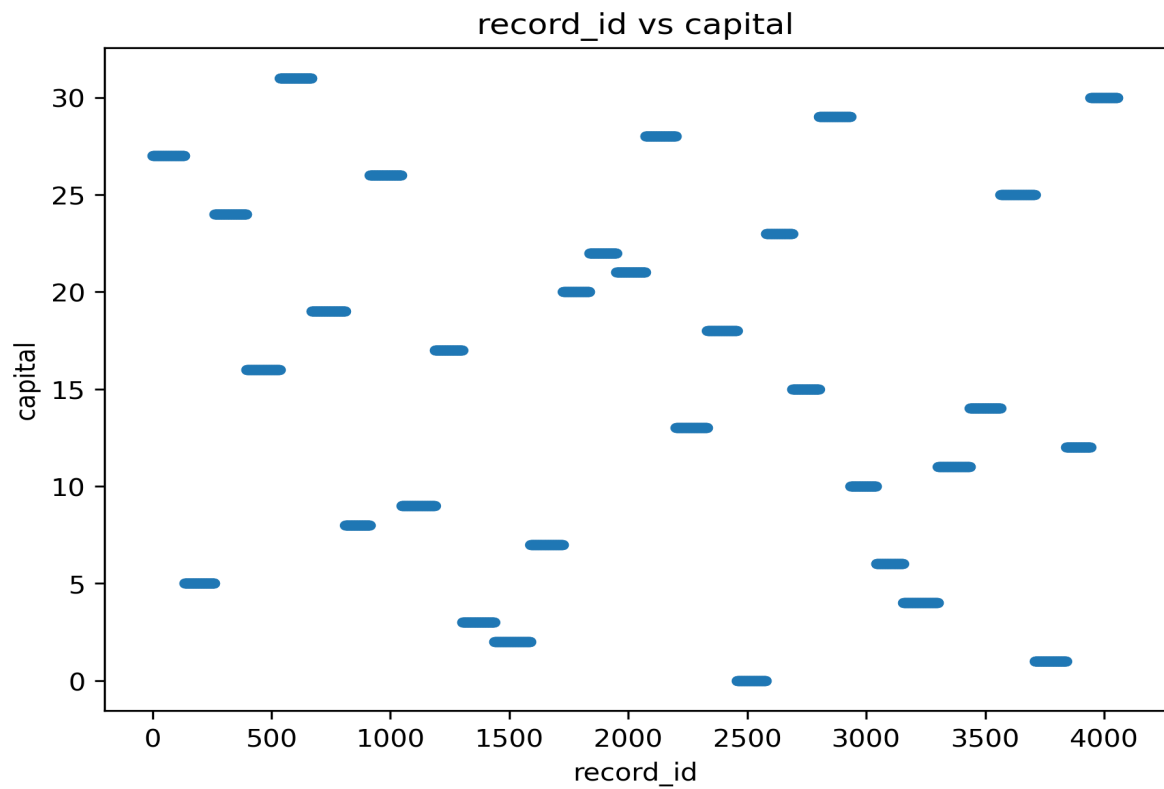
Scatter relationship between record_id and founding_member.



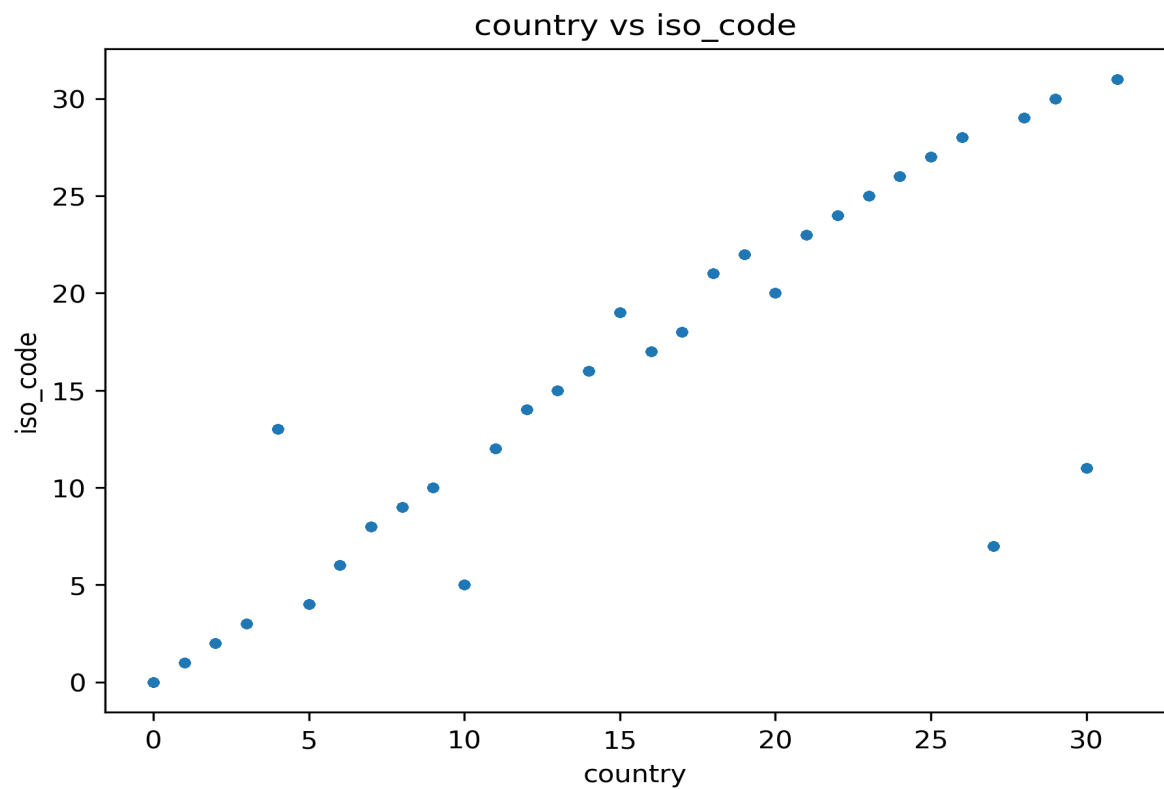
Scatter relationship between record_id and nuclear_sharing.



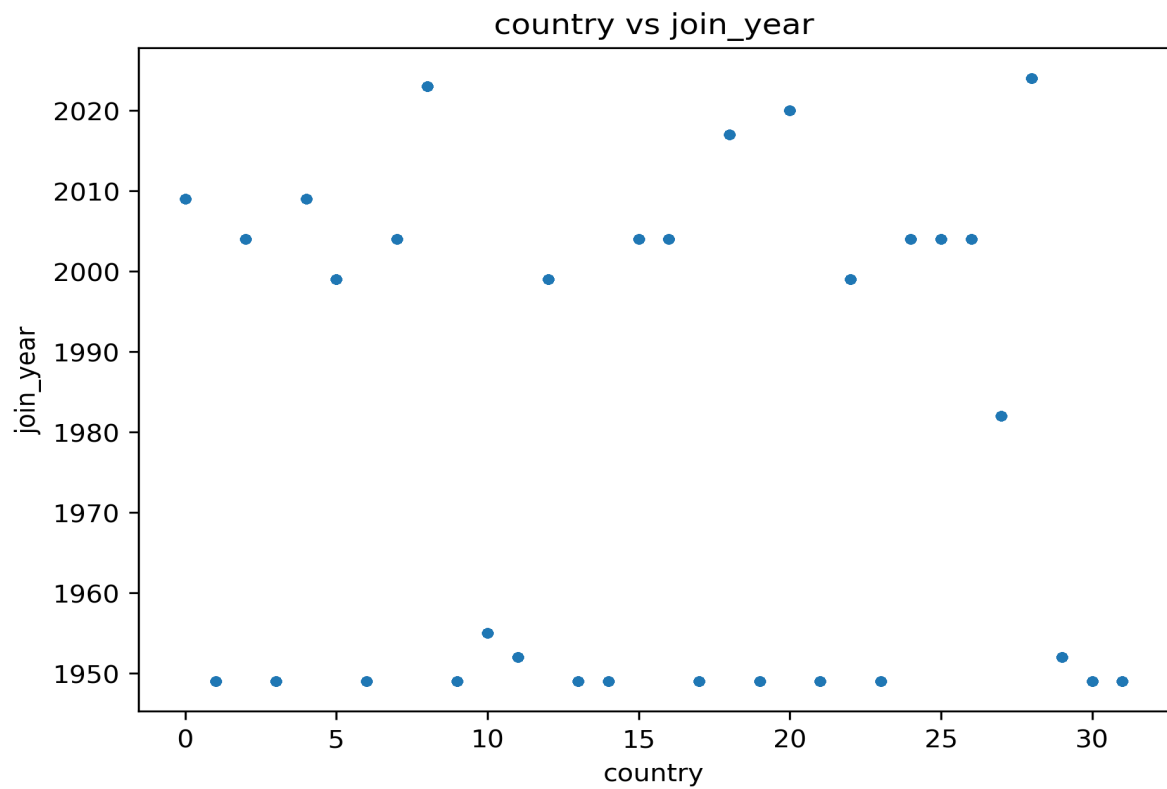
Scatter relationship between record_id and region.



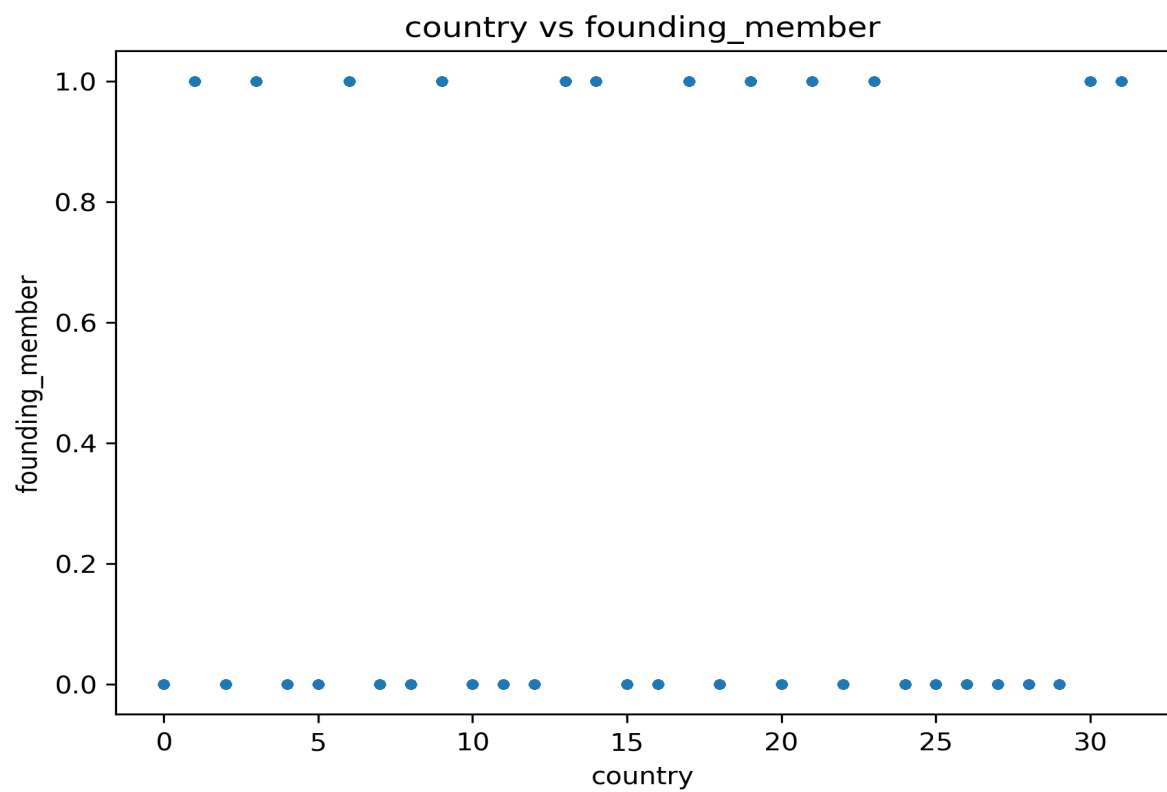
Scatter relationship between record_id and capital.



Scatter relationship between country and iso_code.



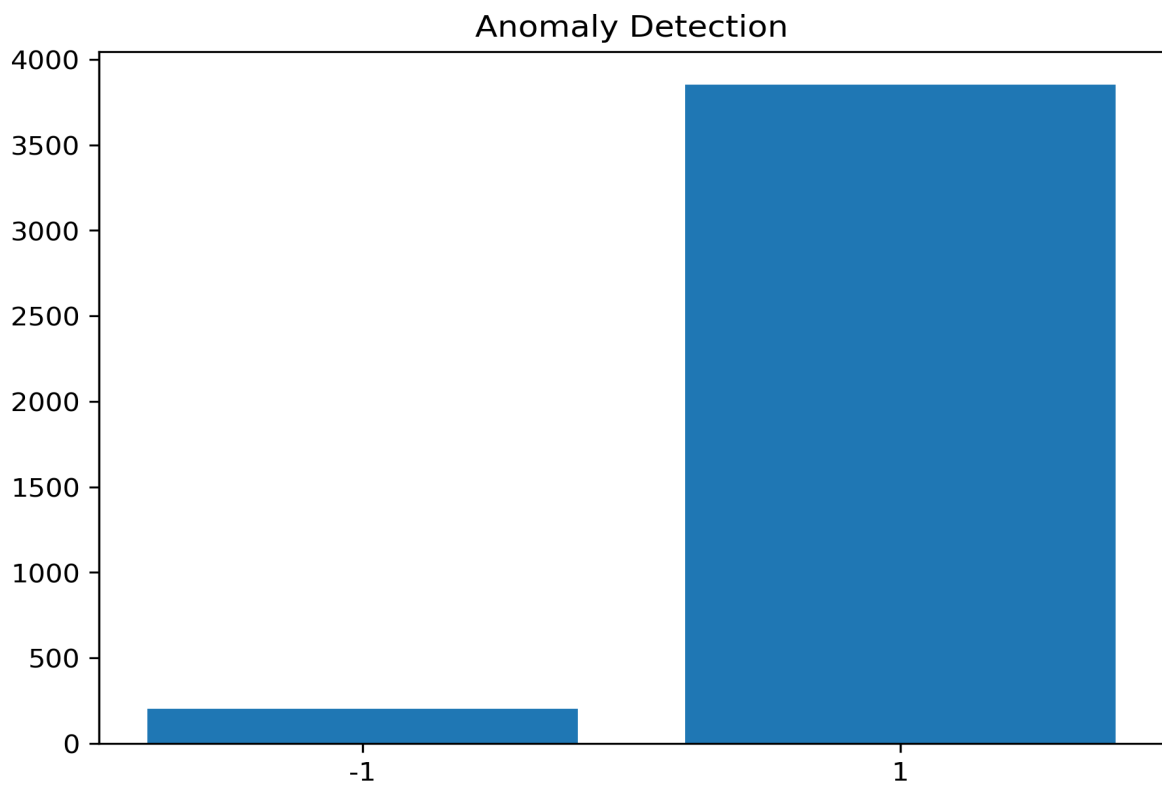
Scatter relationship between country and join_year.



Scatter relationship between country and founding_member.



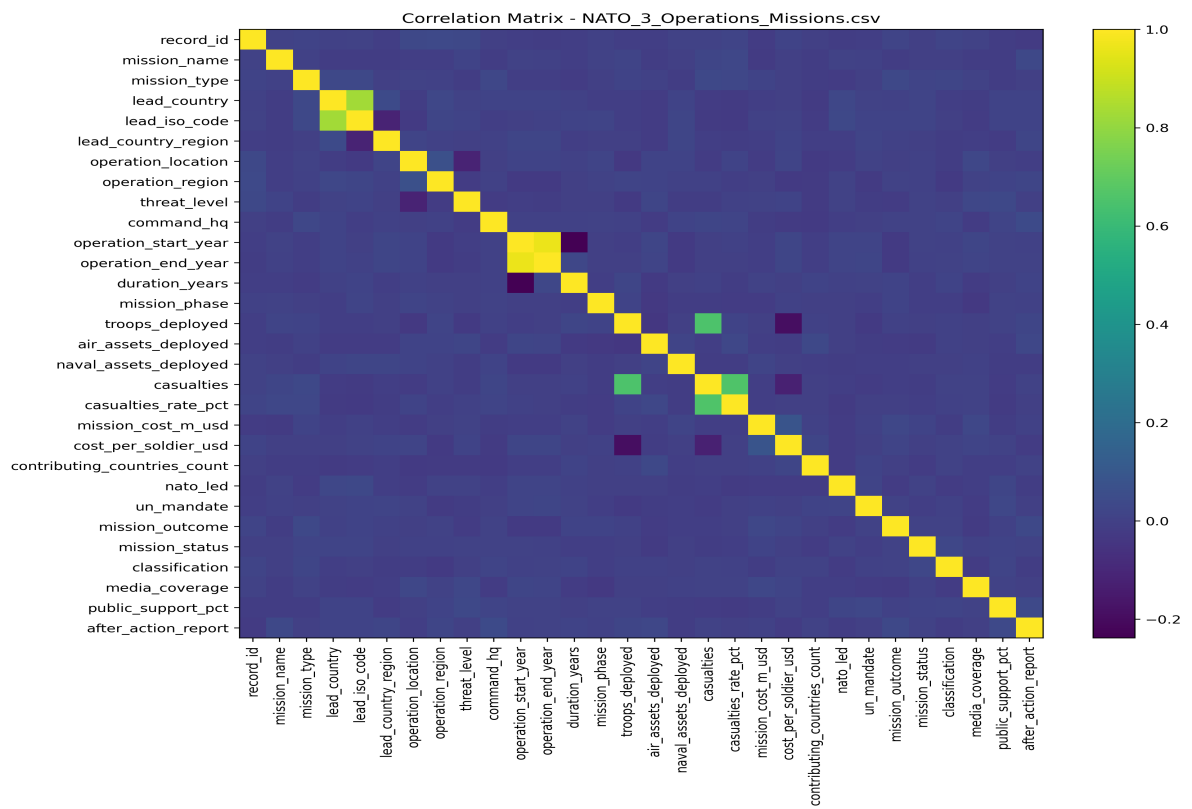
KMeans clustering groups similar records into data-driven clusters.



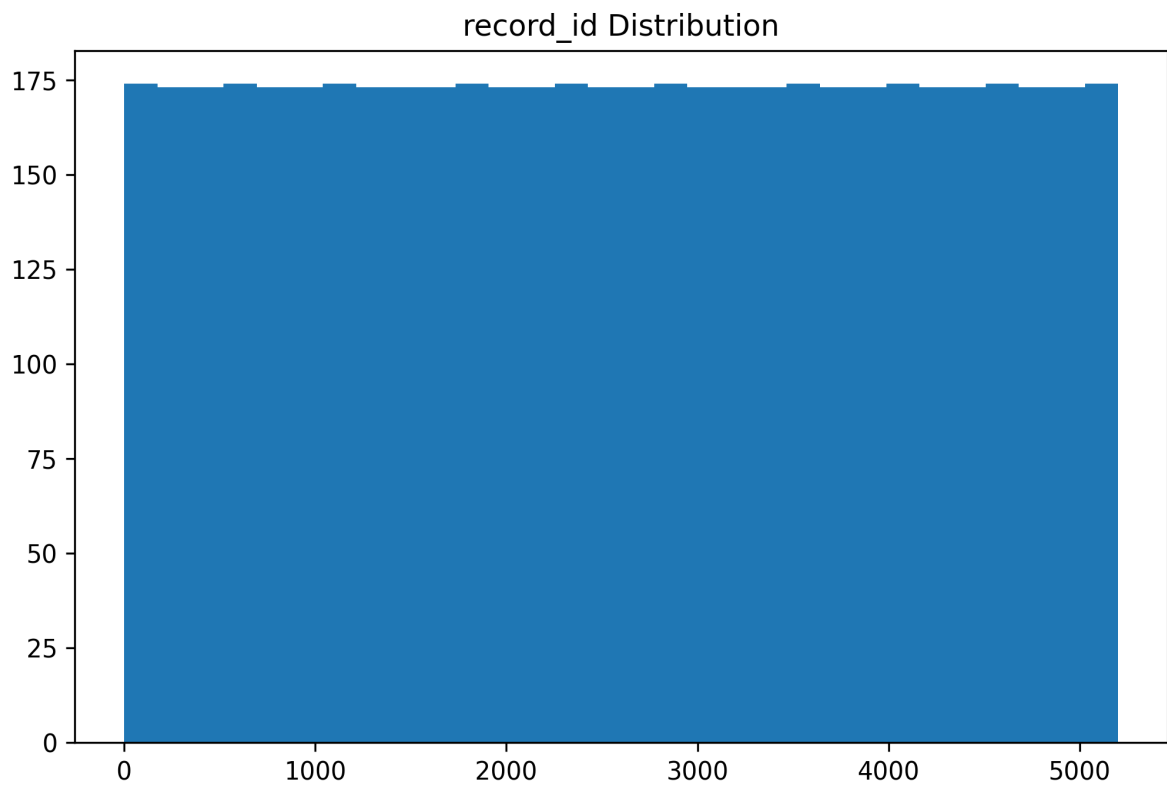
Isolation Forest identifies unusual or abnormal observations within the dataset.

Dataset Analysis: NATO_3_Operations_Missions.csv

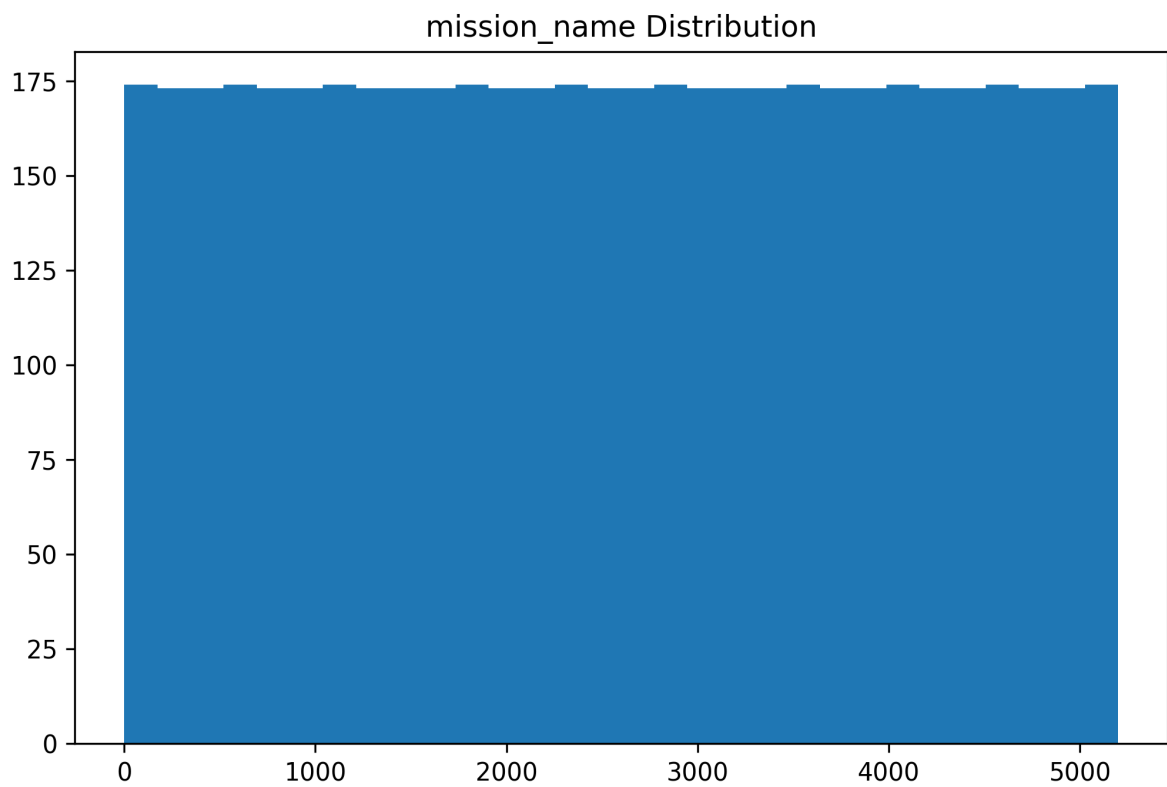
Rows: 5200
Columns: 32
Target Variable: mission_cost_m_usd
Numerical Features: 30
Generated Charts: 21
Random Forest RMSE: 370.657
Gradient Boosting RMSE: 516.907



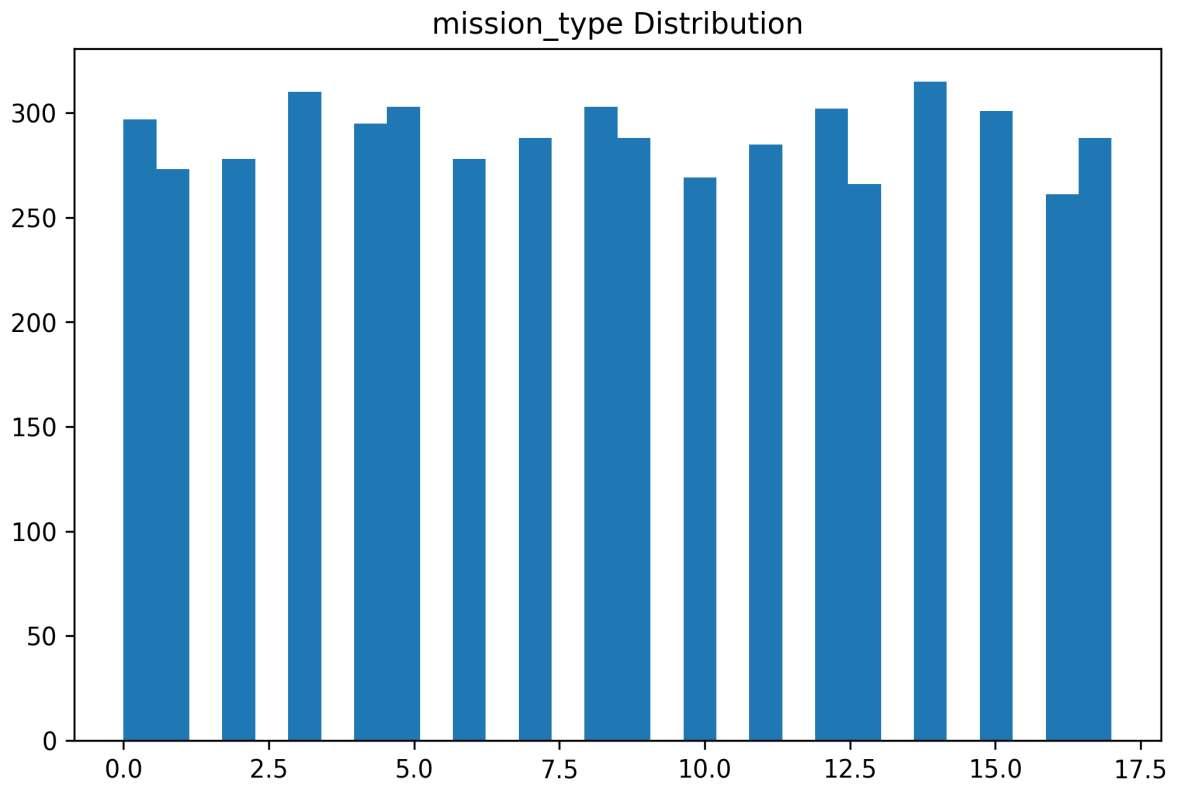
Correlation analysis between numerical variables showing positive and negative feature relationships.



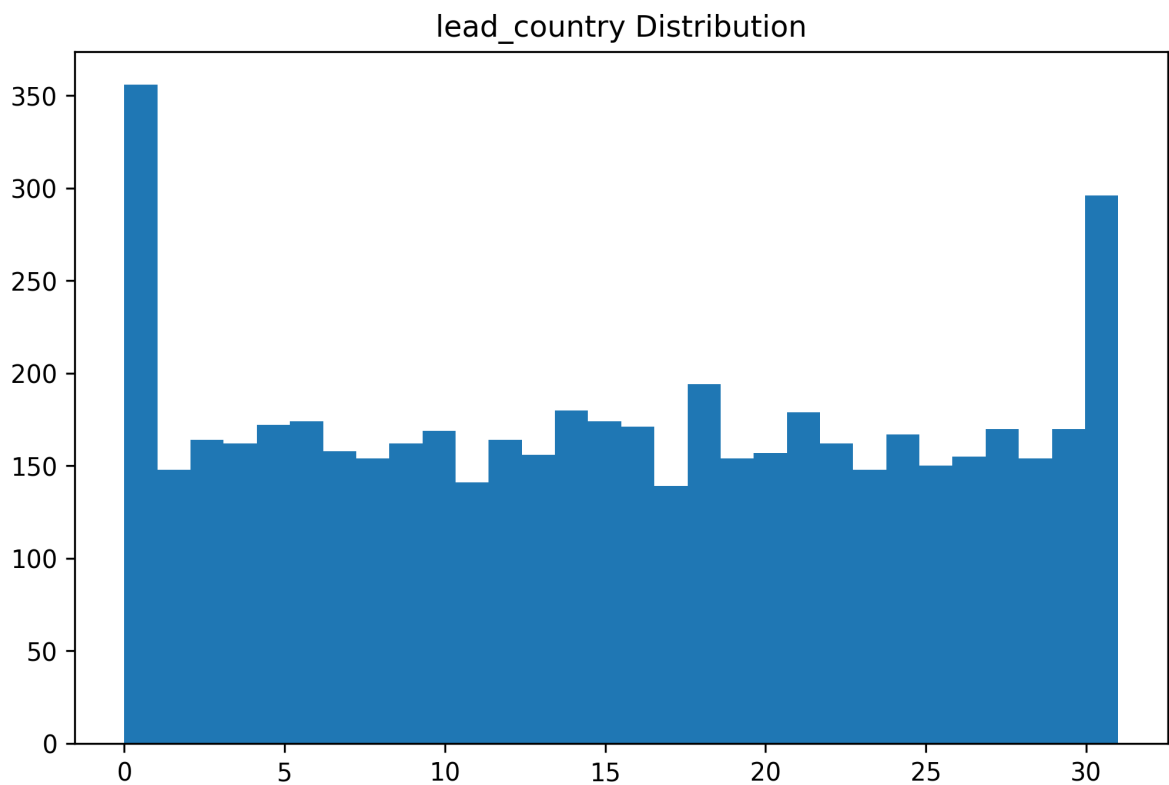
Distribution analysis for record_id showing value spread and density patterns.



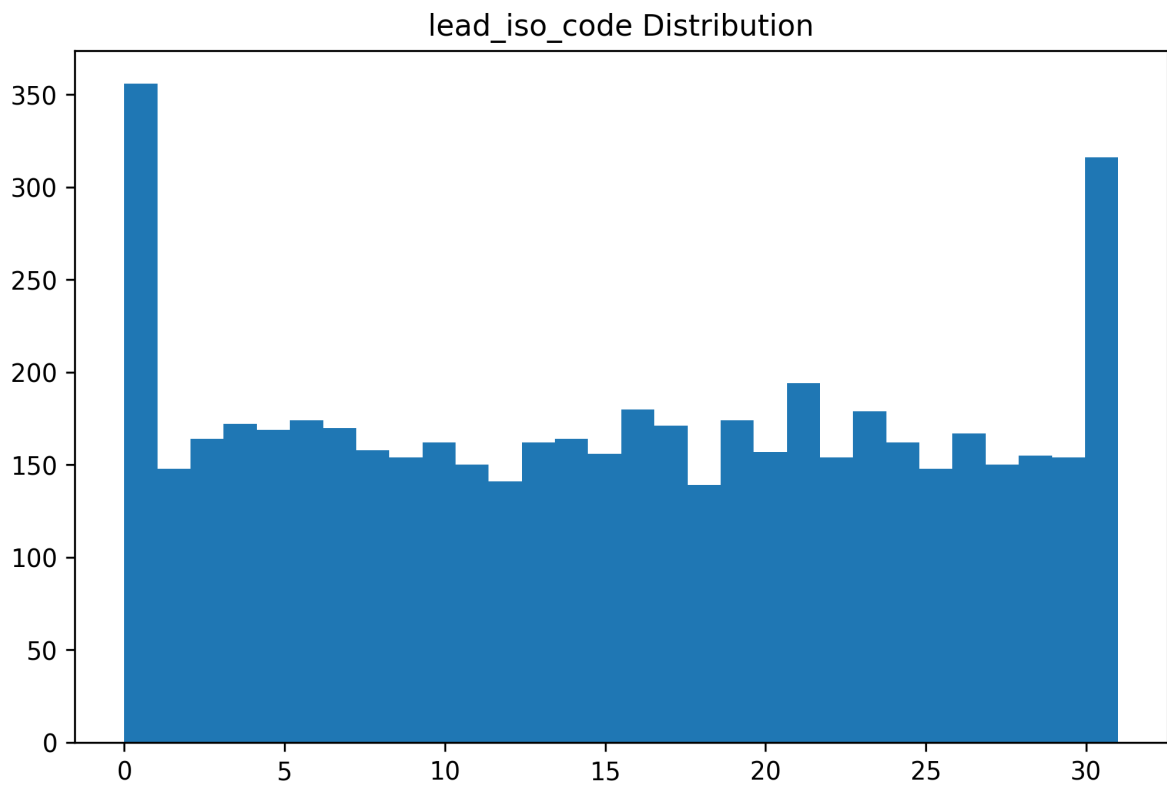
Distribution analysis for mission_name showing value spread and density patterns.



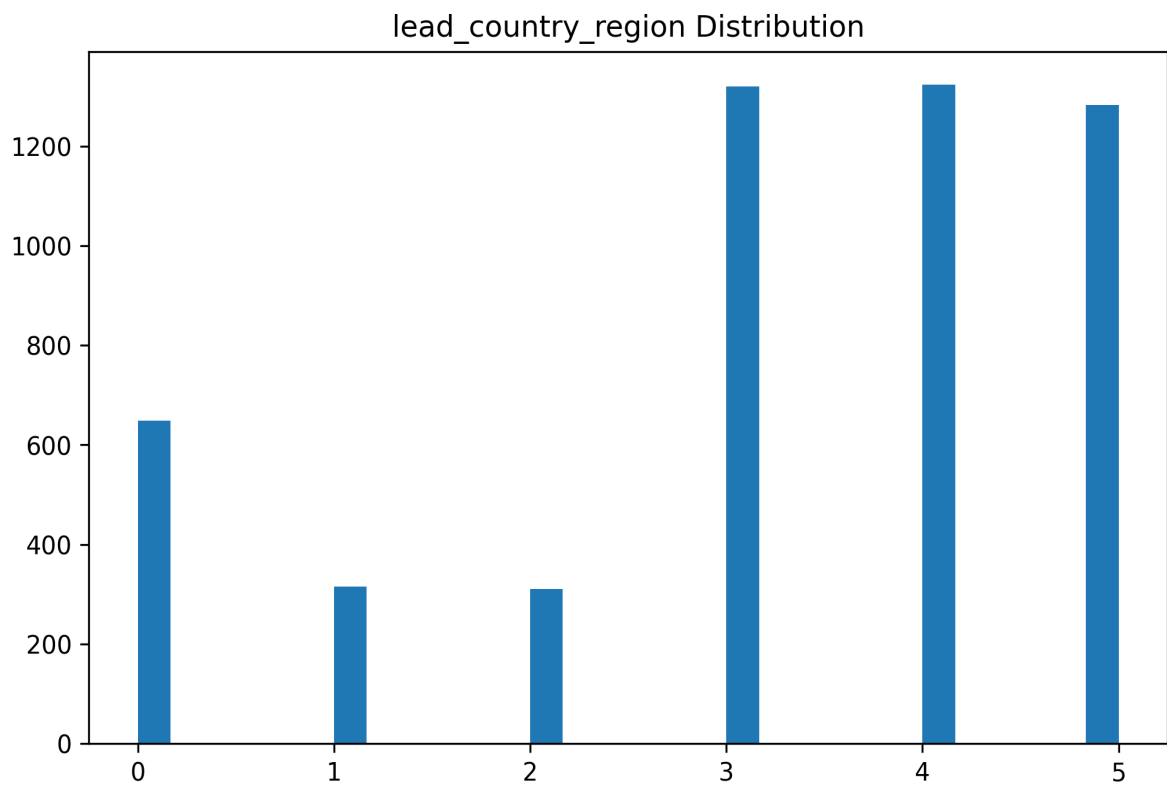
Distribution analysis for mission_type showing value spread and density patterns.



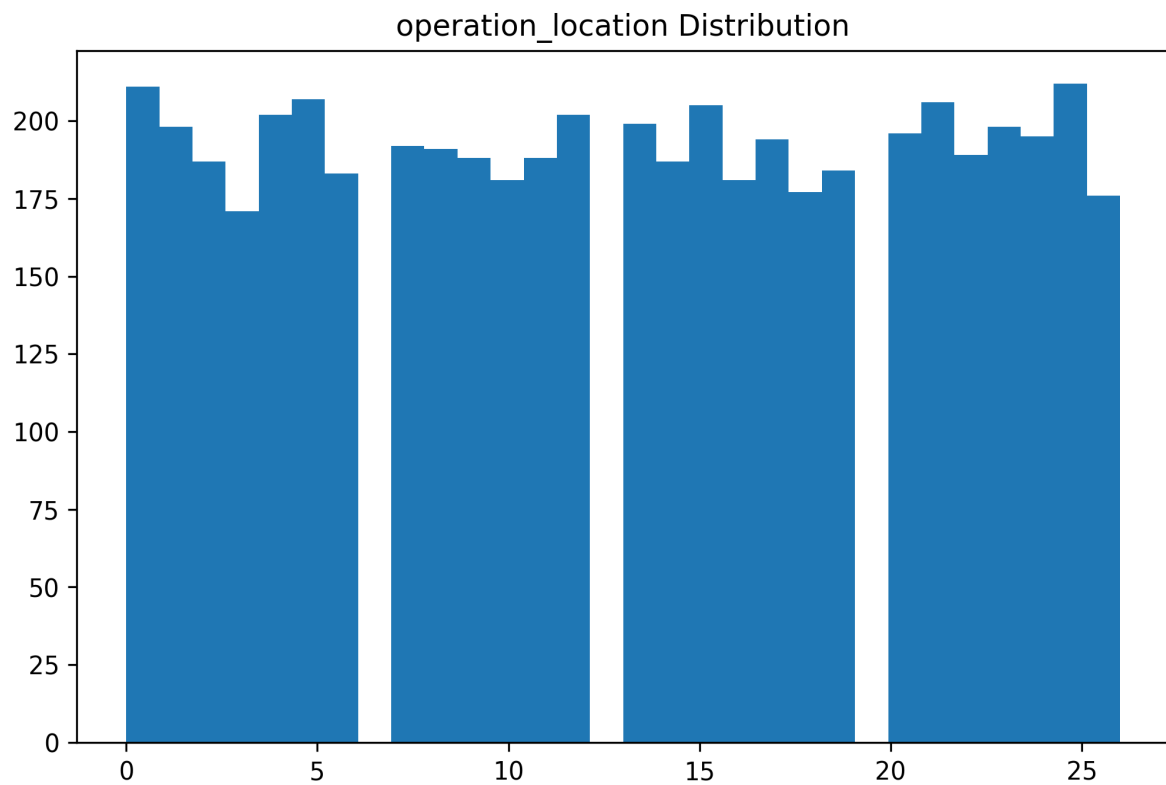
Distribution analysis for lead_country showing value spread and density patterns.



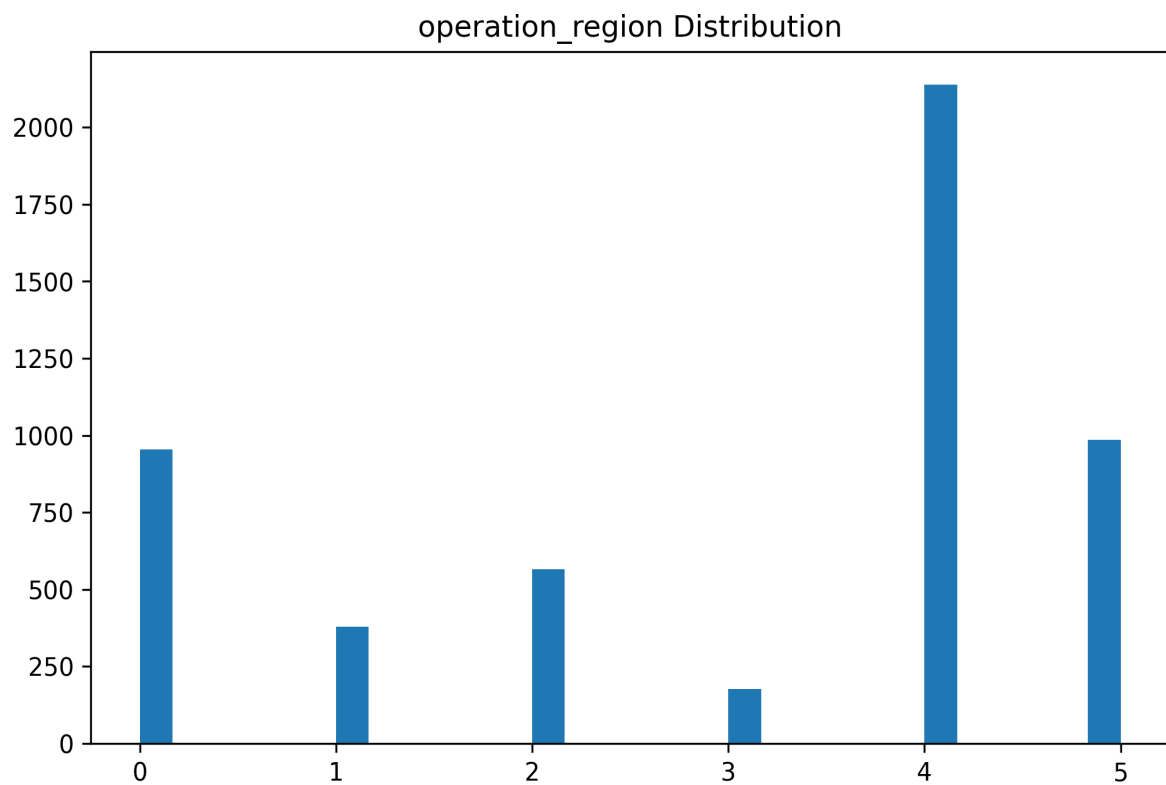
Distribution analysis for lead_iso_code showing value spread and density patterns.



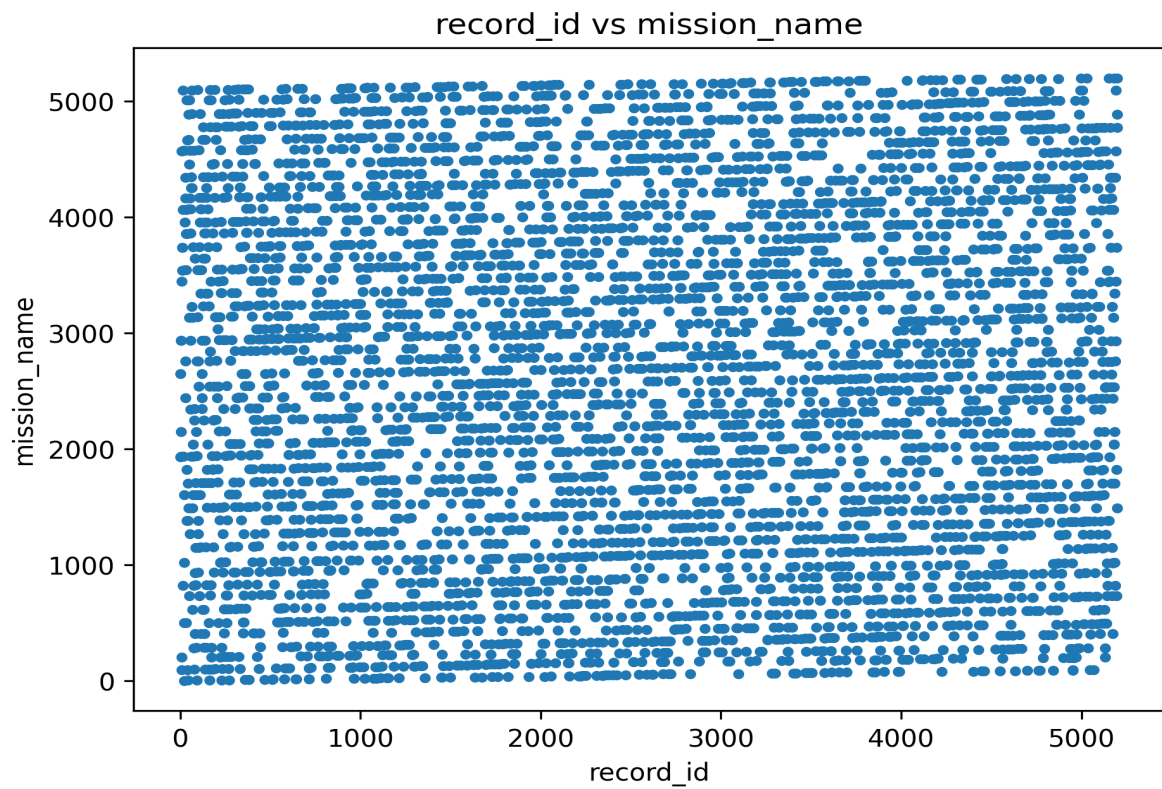
Distribution analysis for lead_country_region showing value spread and density patterns.



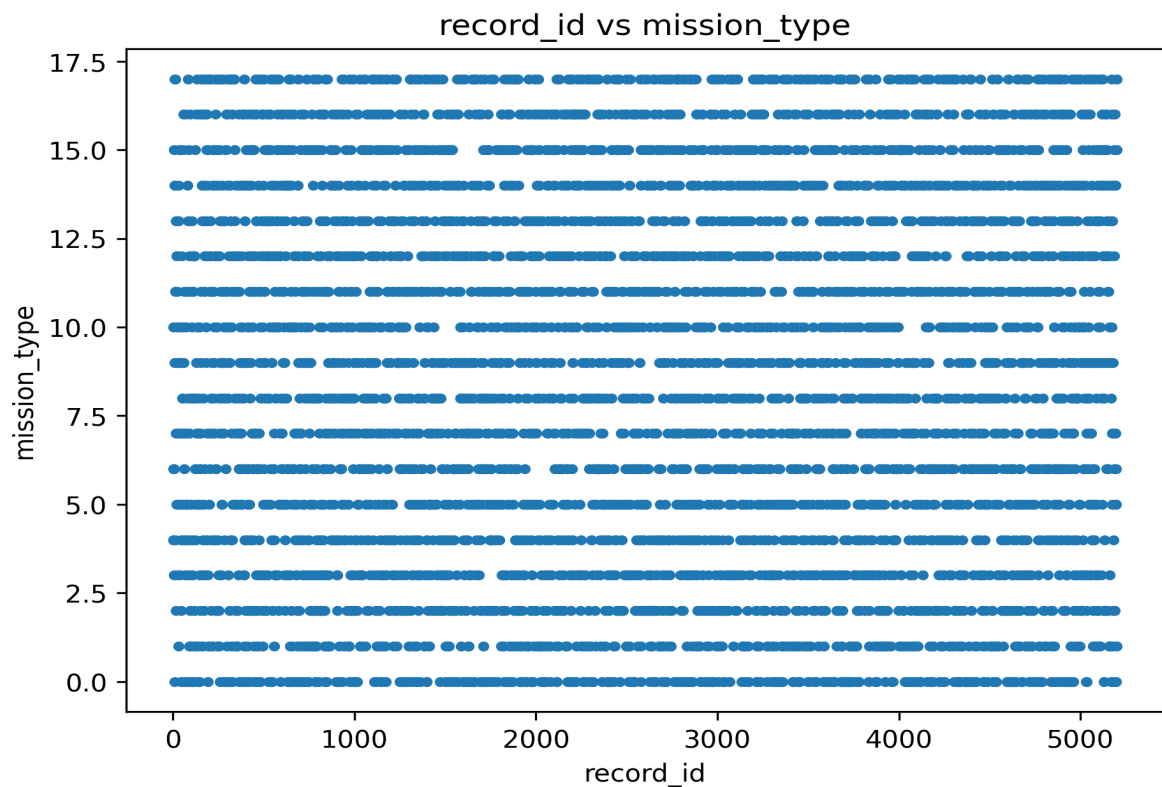
Distribution analysis for operation_location showing value spread and density patterns.



Distribution analysis for operation_region showing value spread and density patterns.



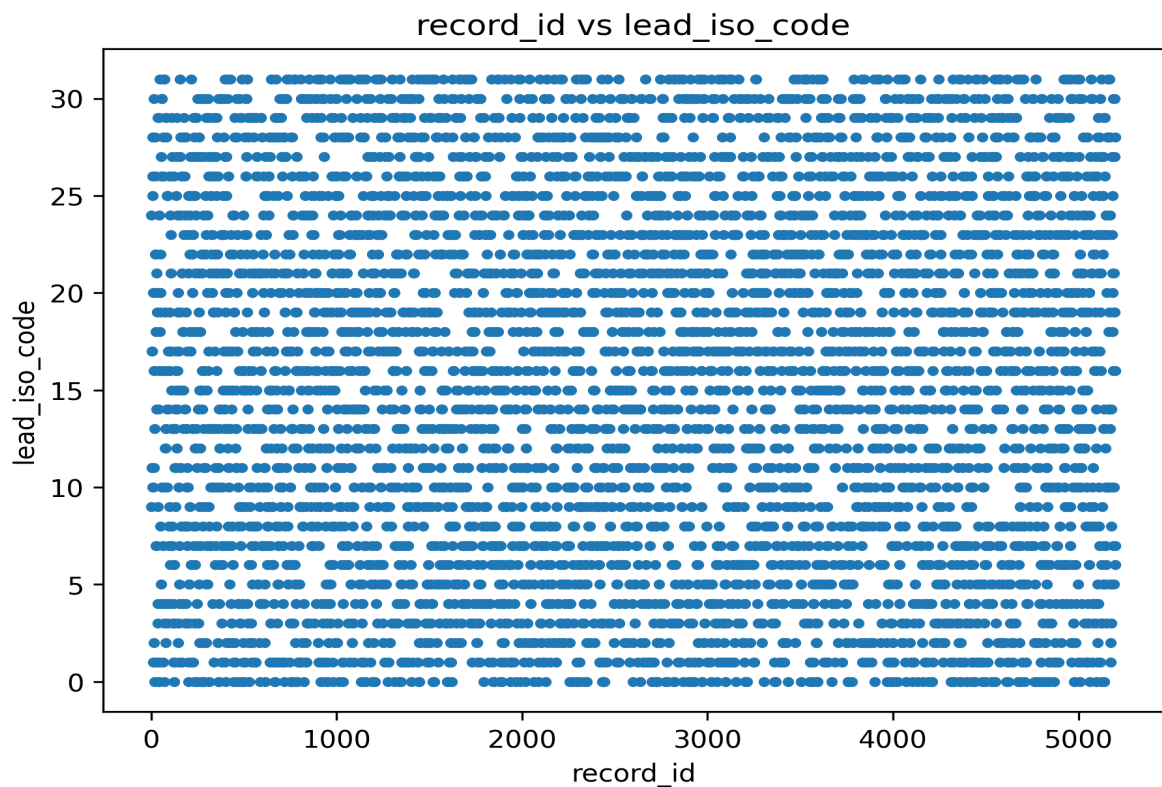
Scatter relationship between record_id and mission_name.



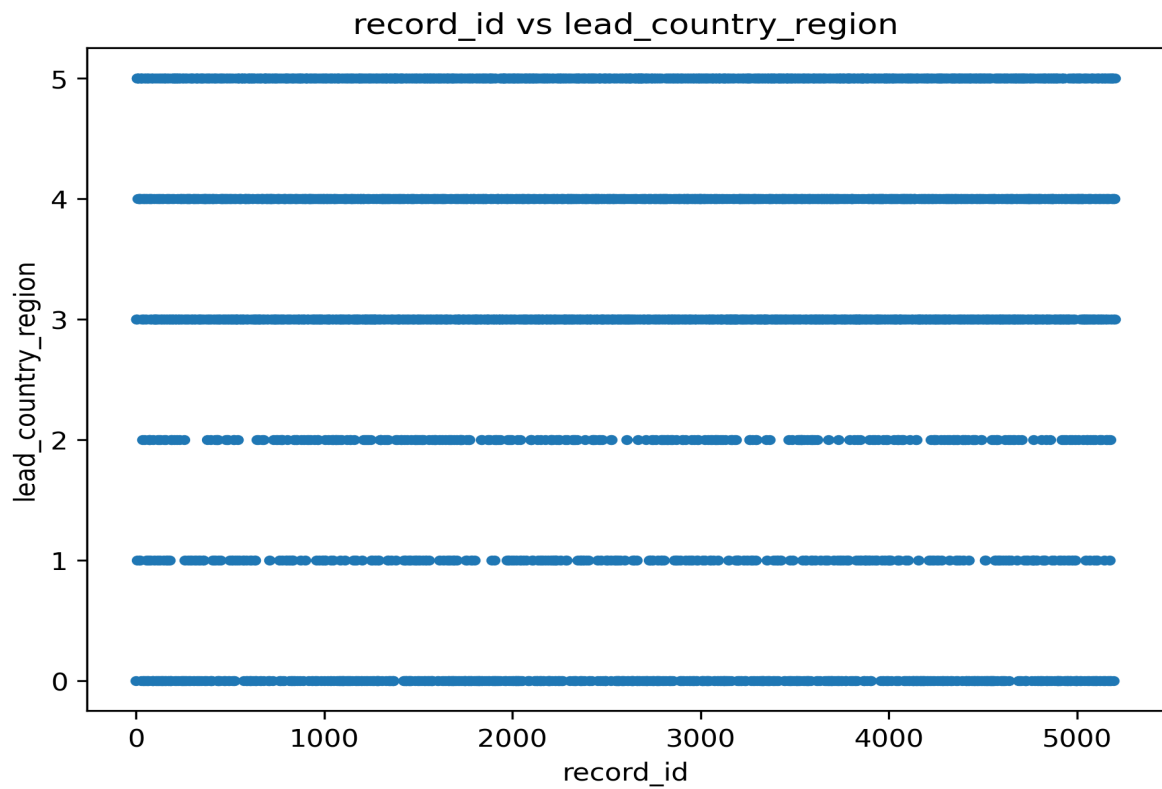
Scatter relationship between record_id and mission_type.



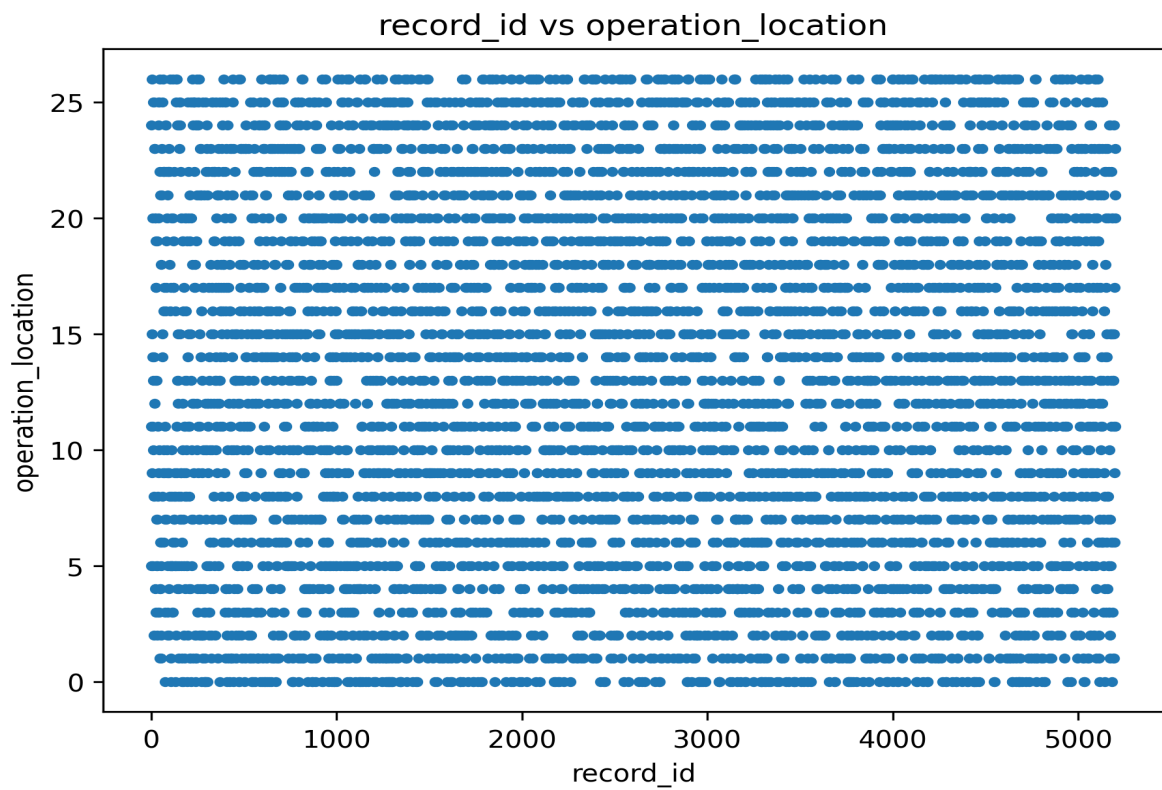
Scatter relationship between record_id and lead_country.



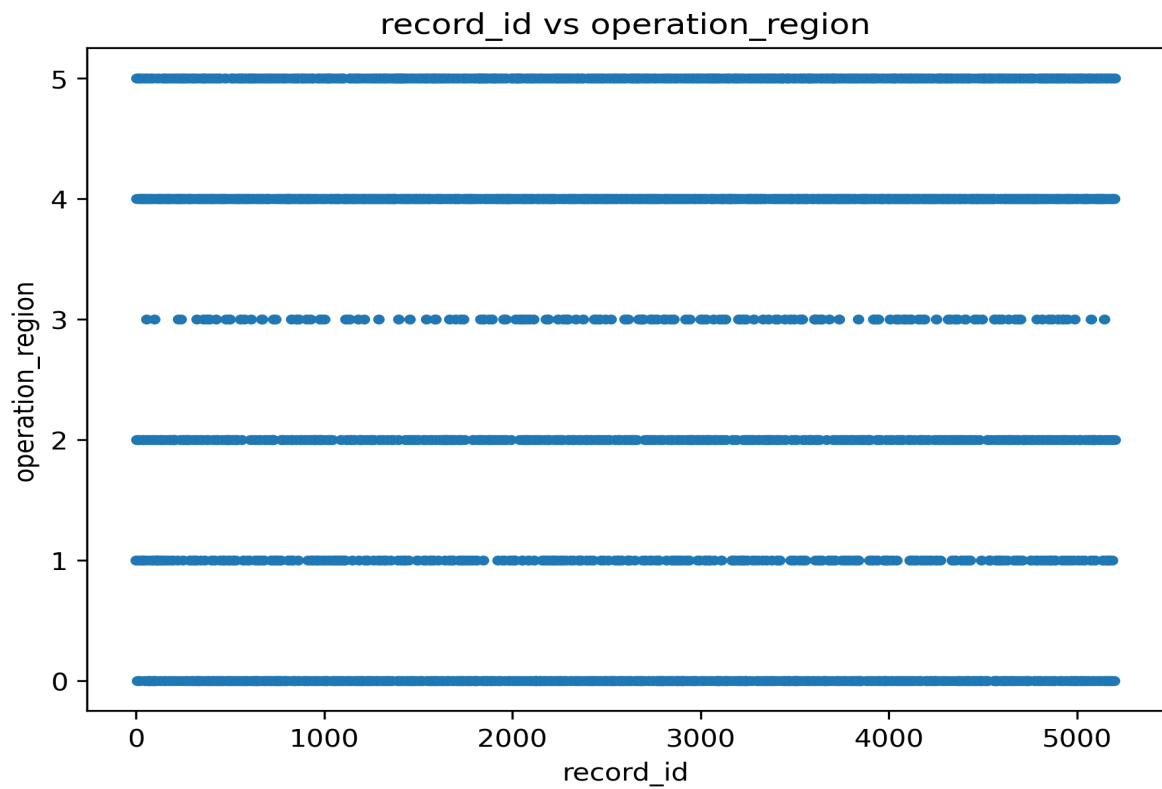
Scatter relationship between record_id and lead_iso_code.



Scatter relationship between record_id and lead_country_region.



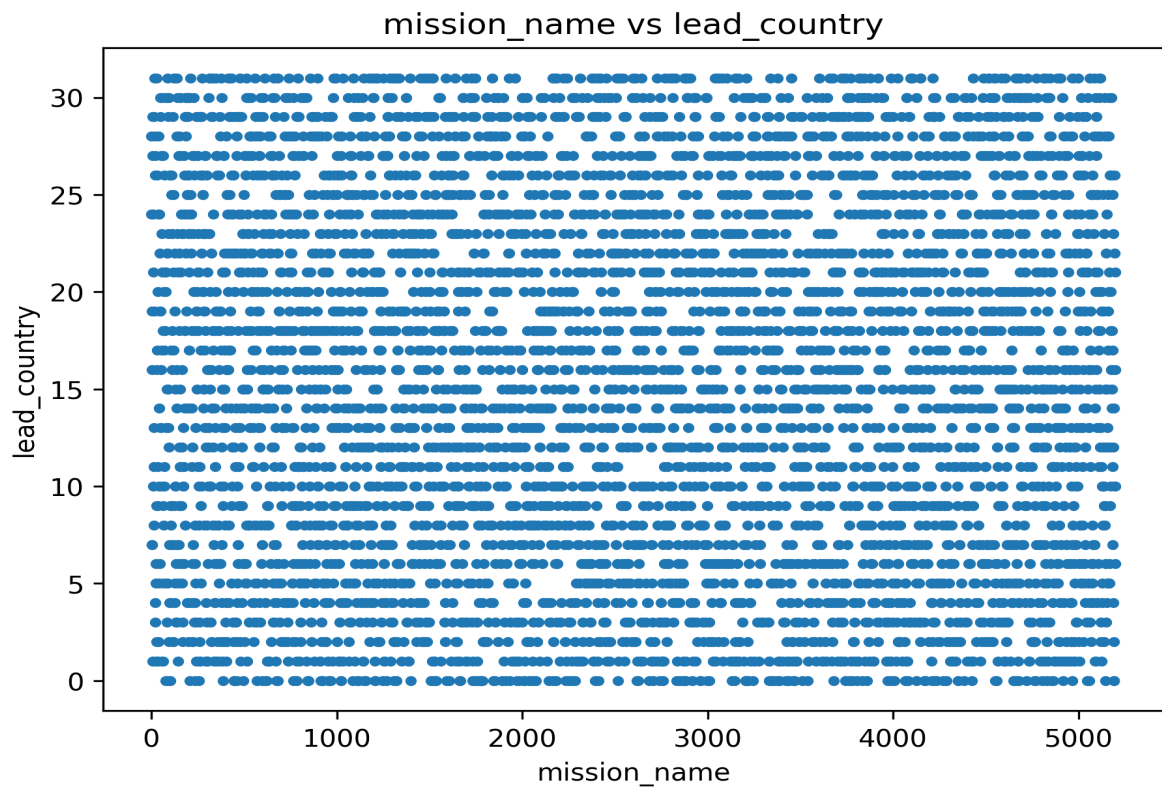
Scatter relationship between record_id and operation_location.



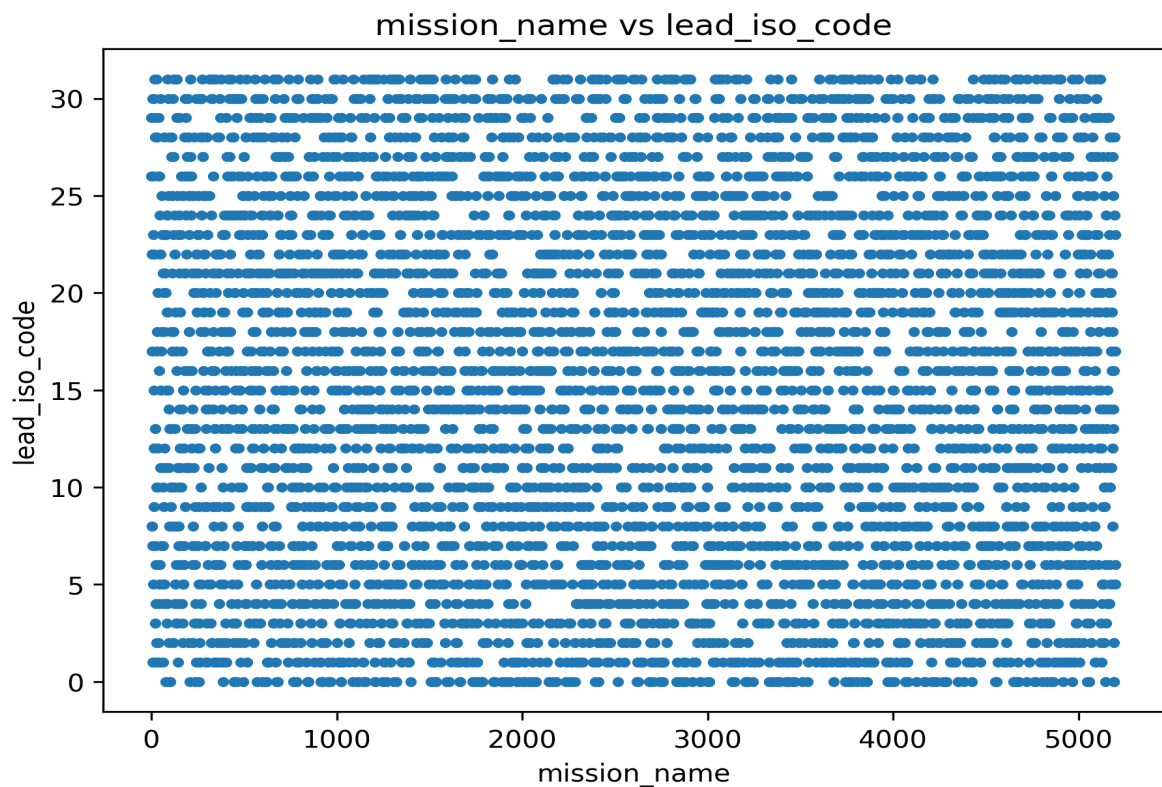
Scatter relationship between record_id and operation_region.



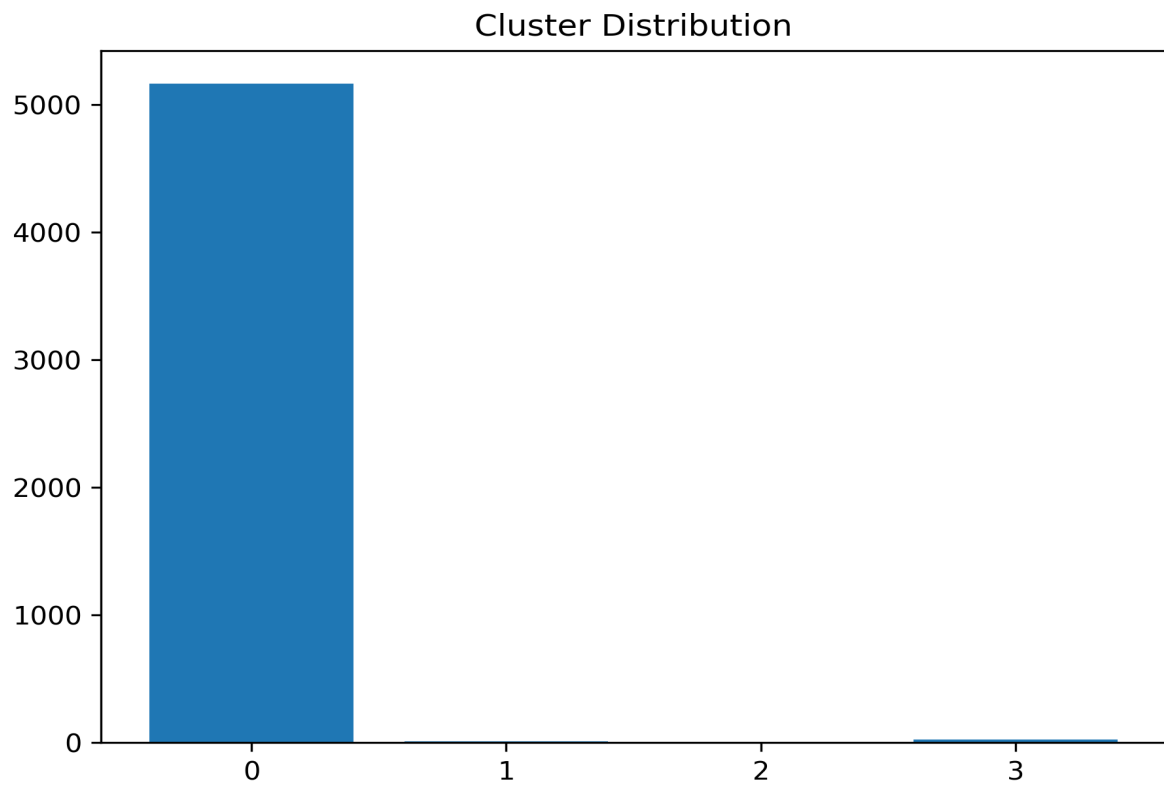
Scatter relationship between mission_name and mission_type.



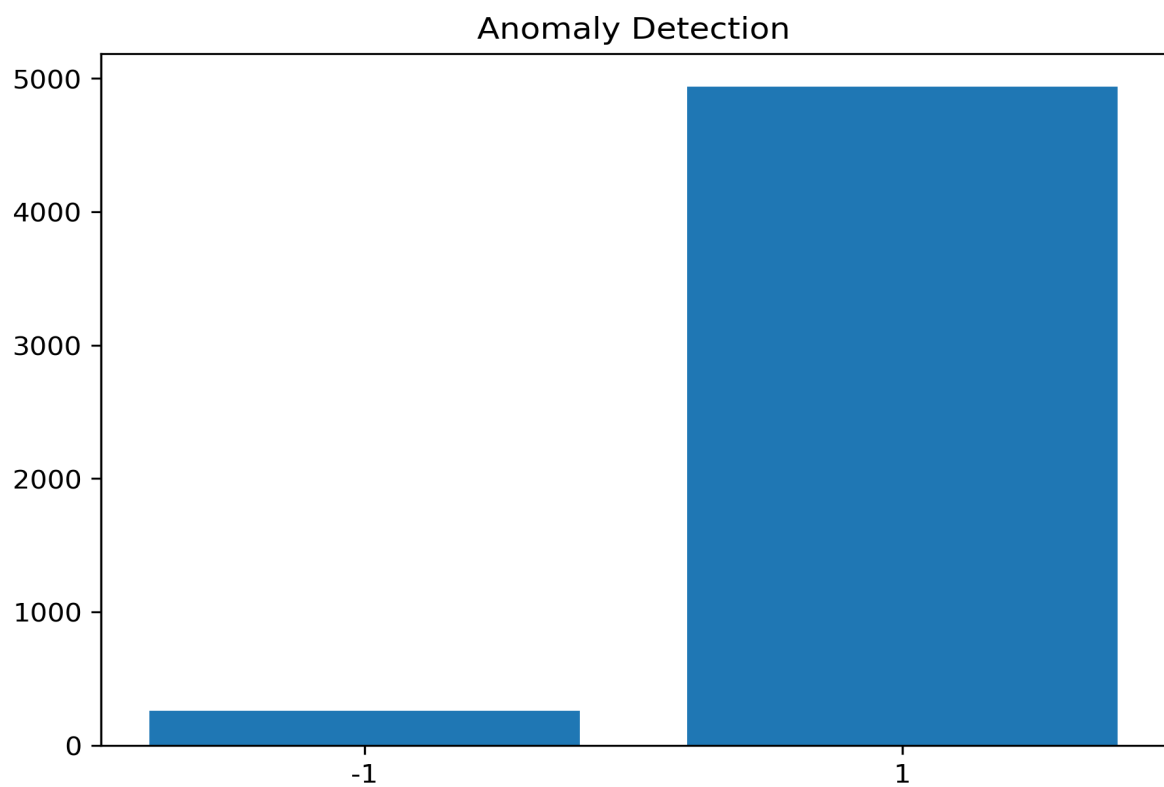
Scatter relationship between mission_name and lead_country.



Scatter relationship between mission_name and lead_iso_code.



KMeans clustering groups similar records into data-driven clusters.



Isolation Forest identifies unusual or abnormal observations within the dataset.

Final Dataset Summary

Processed Datasets:

- NATO_1_Country_Stats.csv

Rows: 1444 | Columns: 29 | Target: gdp_billion_usd | Charts: 21

- NATO_2_Equipment_Inventory.csv

Rows: 4056 | Columns: 27 | Target: units_count | Charts: 21

- NATO_3_Operations_Missions.csv

Rows: 5200 | Columns: 32 | Target: mission_cost_m_usd | Charts: 21